SM-A146P/U/U1/W/S146VL
Common

BB

RF

Title	001_Index	REV:	V10
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DEPARTMENT:	BB	DESIGNER:	liufenglei
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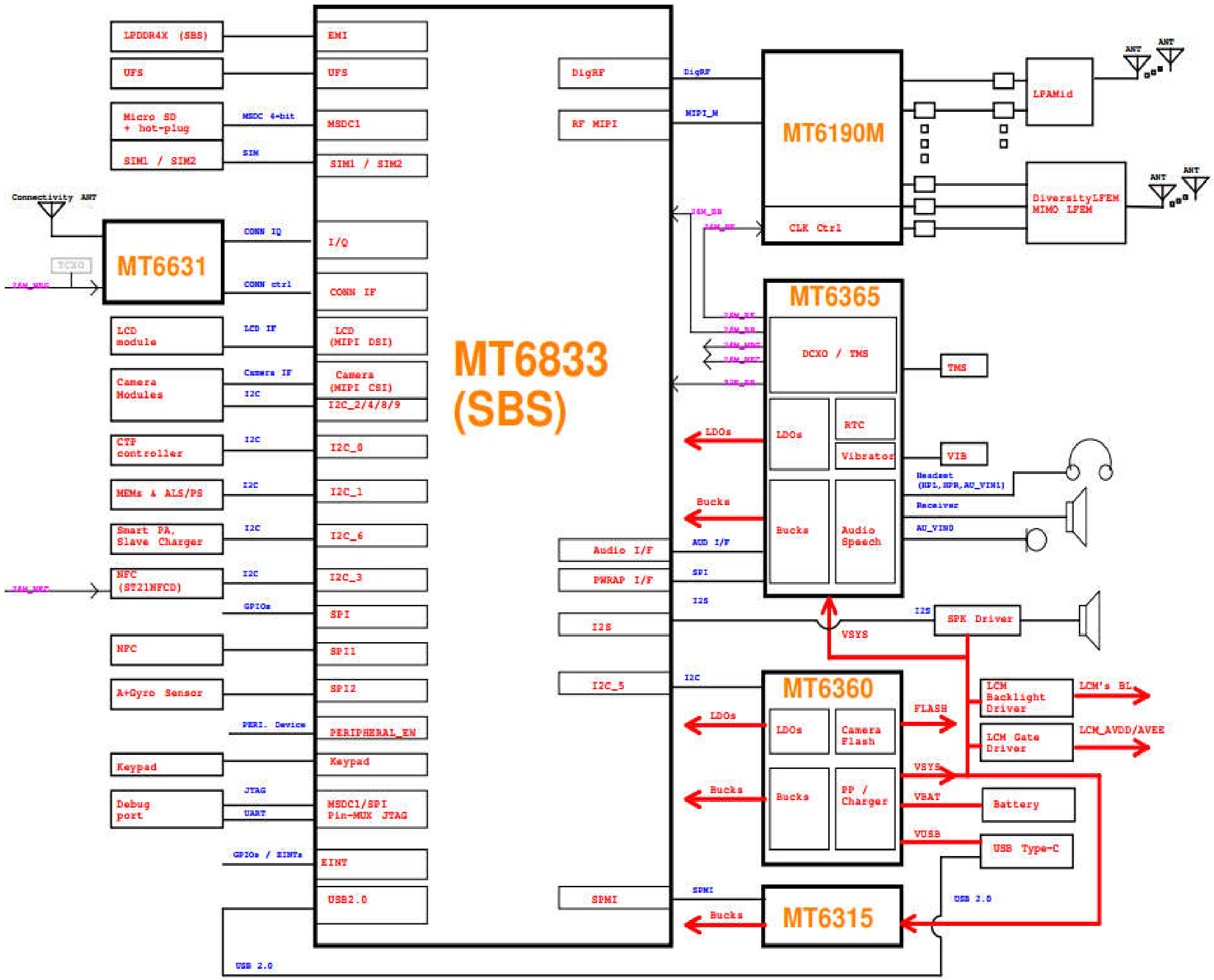
02_Block_Diagram



Title	002_Block_Diagram		REV: V10
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DEPARTMENT:	BB	DESIGNER:	liufenglei
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03_I2C_ID_Overview

I2C	AP, SCP, SSPM	Function	I2C/I3C Spec.	I2C Slave Address (7-bit mode)
I2C-0			I2C	
I2C-1	SCP	M Sensor	I2C	MMC5603NJL I2C address: 0x0C (Read:0x61 Write:0x60)
		ALS+PS Sensor		AK09918C 0x46 I2C address: 0x46 (Read:0x19 Write:0x18)
I2C-2	AP	Main CAM **M	I2C	Hi1336： I2C address :0x1A (Read:0x41 Write:0x40) ； OV13B： I2C address :0x6C (Read:0x6D Write:0x6C) ； IMX258 I2C address: 0x50 (Read:0x21 Write:0x20) S5K3L6 I2C address: 0x50 (Read:0x21 Write:0x20)
I2C-3	AP	NFC	I2C	
		SAR		A96T346DFP I2C Address 0x20 (Read:0x41 Write:0x40)
I2C-4	AP	Front Camera	I2C	S5K4H7 I2C Address : (Read:0x5B Write:0x5A) Hi846 I2C Address： （Read:0x41 Write:0x40) GC8054 2C Address： （Read:0x6F Write:0x6E)
		Macro 2 M CAM		GC2375 I2C Address (Read:0X2F Write:0X2E) OV02A10 I2C Address (Read:0x7B Write:0x7A)
I2C-5	AP	MT6360	I2C	MT6360 PD's I2C address: 0X4E (Read:0x9D Write:0x9C) MT6360 PMIC's I2C address: 0x1A (Read:0x35 Write:0x34) MT6360 PMU's I2C address: 0x34 (Read:0x69 Write:0x68) MT6360 LDO's I2C address:0x64 (Read:0xC9 Write:0xC8)
I2C-6	AP	Smart PA	I2C	AW88264CSR Speaker AMP I2C Address: 0x34(Read:0x69 Write:0x68)
		LCM Gate Driver		SM5109 I2C address:0X3E (Read:0x7D Write:0x7C)
		LCM BACKLINGHT		AW99703 I2C address:0x36 (Read:0x6D Write:0x6C)
I2C-7	AP		I2C	TBD
I2C-8	AP	Depth 2 M CAM	I2C	OV02B1B I2C address: (Read:0x78 Write:0x79)
				GC2375B I2C address: (Read:0x2F Write:0x2E)
I2C-9	AP		I2C	

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

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04_Change_Notice

Date	Version	Page	Change notice
20120.10.15			Changed from 5G-C HW
20120.10.16			1st release

Title	004_Change_Notice	REV:	V10
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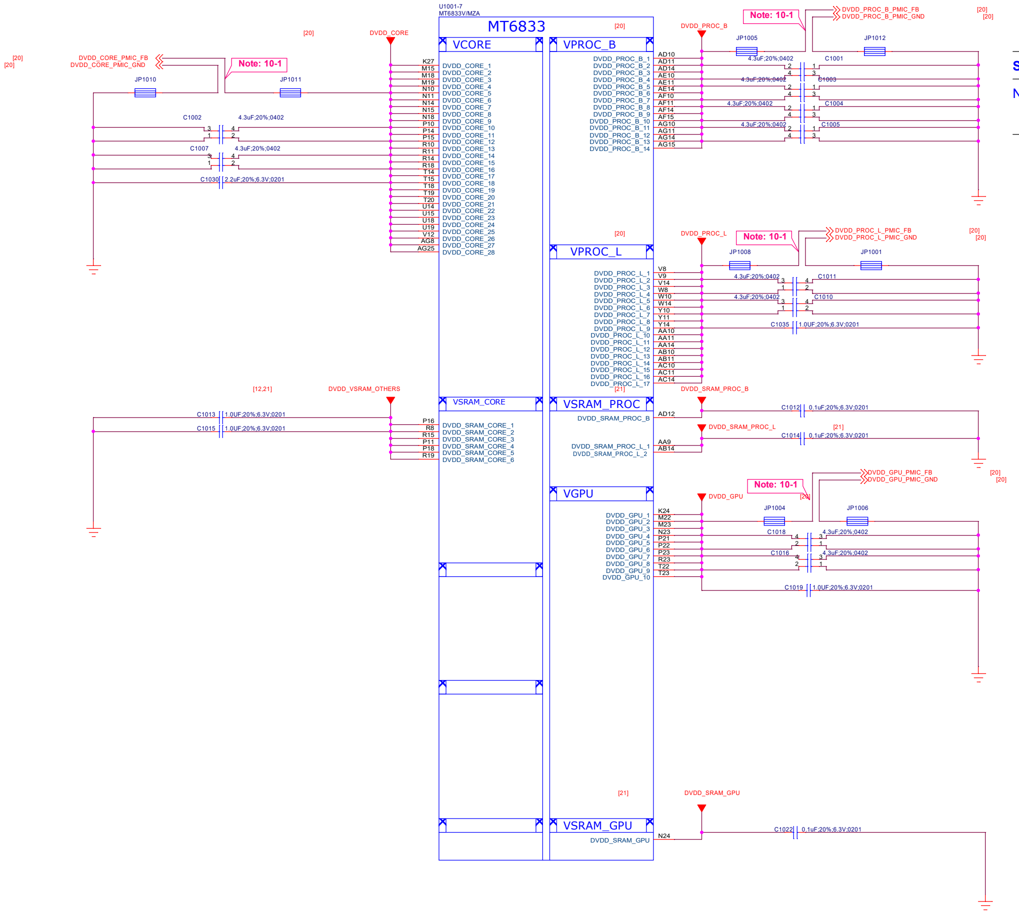
10_BB_POWER_PDN



Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1: Differential pair of PMIC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Title	010_BB_POWER_PDN	REV:	V10
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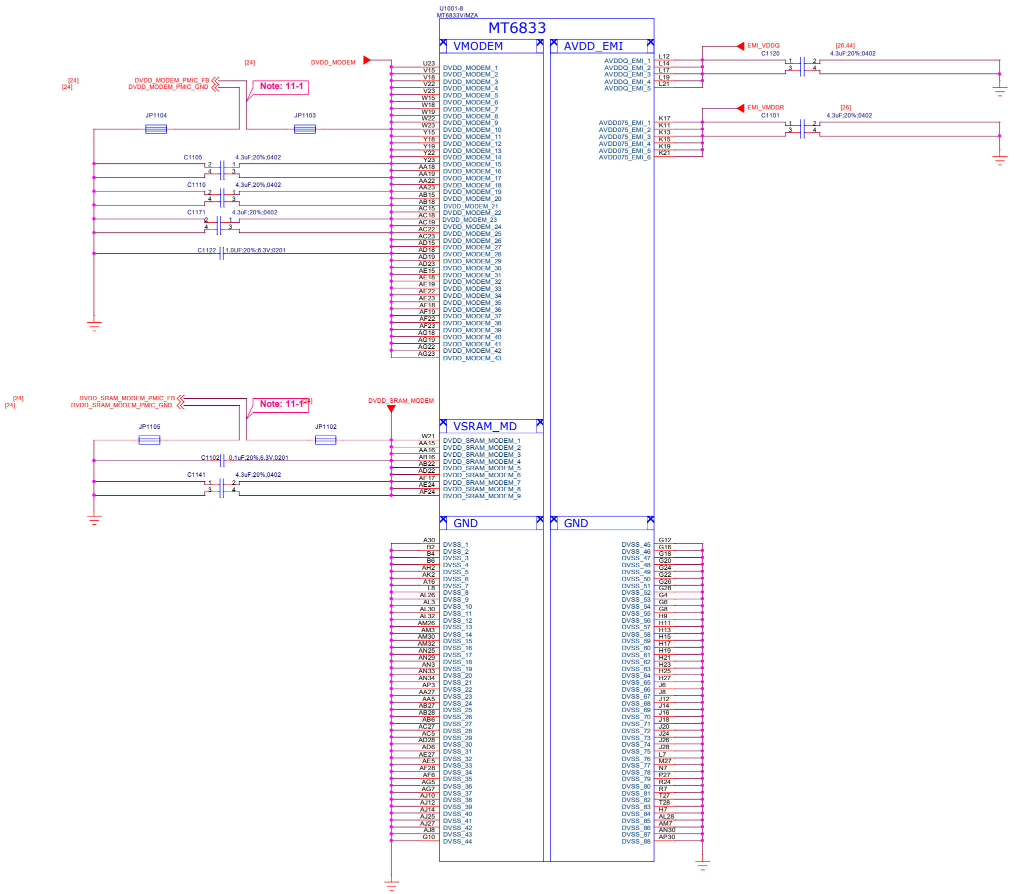
11_BB_POWER_PDN



Schematic design notice of "11_BB_POWER_PDN" page.

Note 11-1: Differential pair of PMIC remote sense must be close to BB's ball.

Remote sense trace with GND shielding to PMIC (Differential)

Title	011_BB_POWER_PDN	REV:	V10
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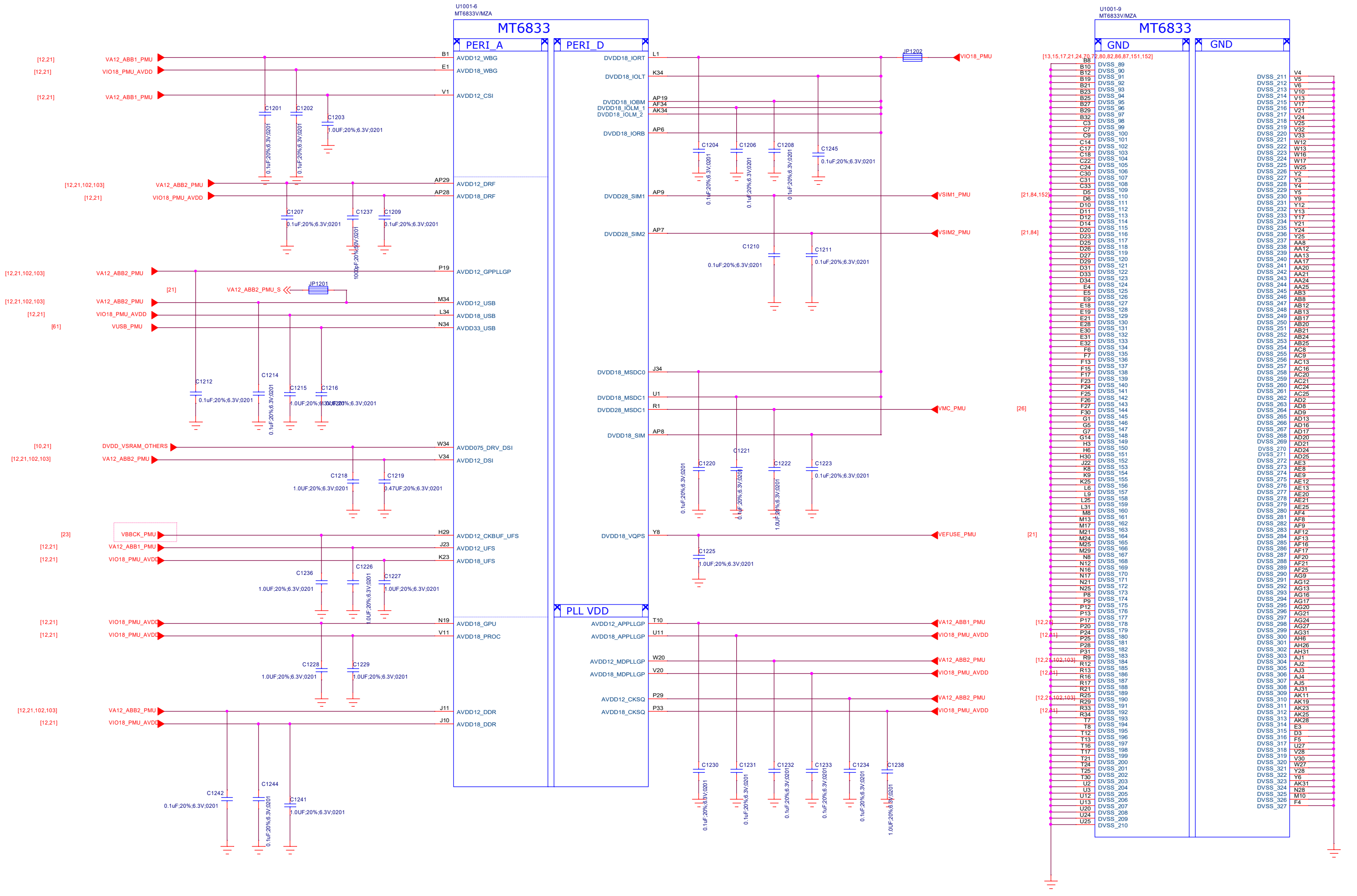
12_BB_POWER_IO



Title	012_BB_POWER_IO	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
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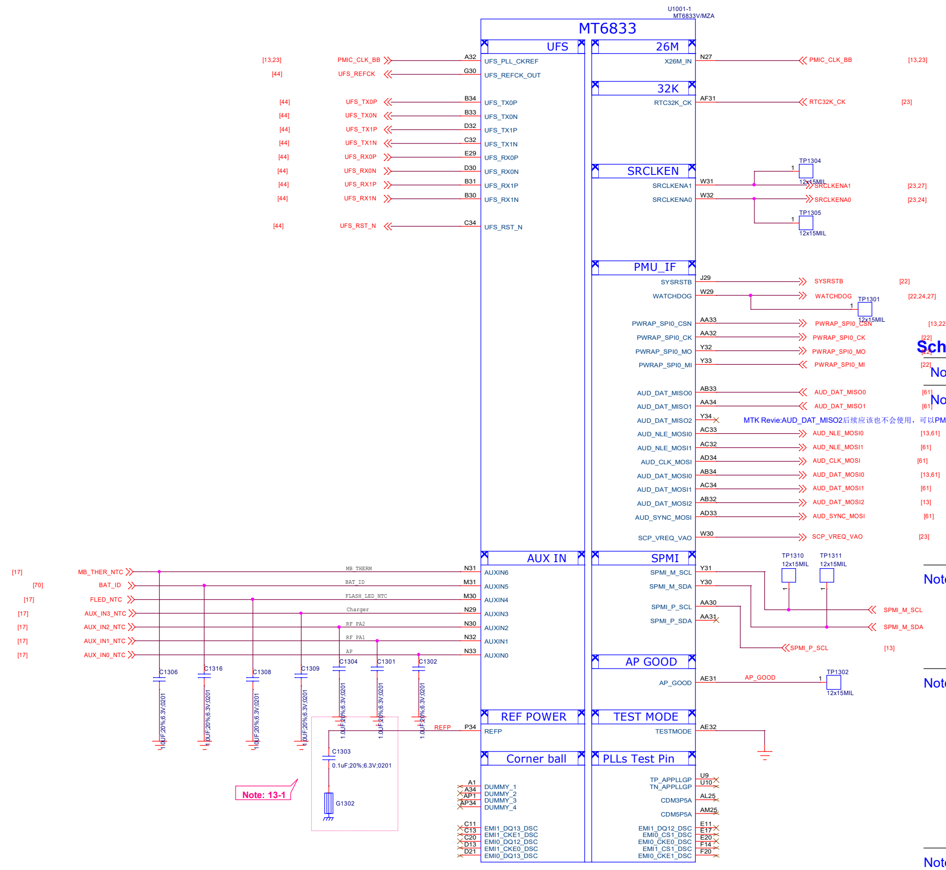
13_BB_1

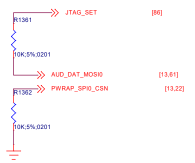


Schematic design notice:

Note 13-1: The load cap. have to be placed as close to REFP ball as possible.

Note 13-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MOS0"I pin features in trapping pin to enable JTAG.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP JTAG	IO JTAG
H (Default)	L (Default)	N/A	N/A
H	H (by external PU)	SPI0+EINT8	N/A
L (by external PD)	L	SPI0+EINT8	EINT16/EINT17/EINT18/EINT19/EINT20 EINT21/EINT22/I2C7
L (by external PD)	H (by external PU)	MSDC1_CLK, MSDC1_CMD, MSDC1_DAT0, MSDC1_DAT1, MSDC1_DAT2	N/A

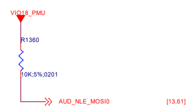


Note 13-3: "AUD_NLE_MOSI1" features in trapping pin to enable JTAG over USB 2.0 port.

AUD_NLE_MOSI1	USB JTAG
L (Default)	USB 2.0
H (by external PU)	DP/DM output JTAG

Note 13-4: "AUD_SYNC_MOSI" and "AUD_NLE_MOSI0" pin features in trapping pin to booting (eMMC/UFS/SPI NOR).

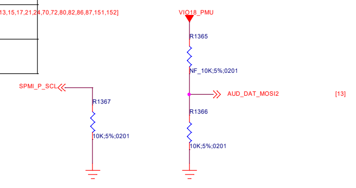
AUD_SYNC_MOSI	AUD_NLE_MOSI0	Storage Booting
L (Default)	L (Default)	Only UFS boot
L	H (by external PU)	Only eMMC boot
H (by external PU)	L	Reserved
H (by external PU)	H (by external PU)	Reserved



Note 13-5: "AUD_DAT_MOSI1" and "AUD_DAT_MOSI2" pin features in trapping pin to enable DDR.

AUD_DAT_MOSI1	AUD_DAT_MOSI2	SPMI_P_SCL	DDR
L (Default)	L (Default)	L (Default)	LP4x-MCP-4266Mbps
L	H (by external PU)	L (Default)	LP4x-Discrete-4266Mbps
H (by external PU)	L	L (Default)	Reserved
H (by external PU)	H (by external PU)	L (Default)	Reserved

Note : SPMI_P_SCL for DDR 4266Mbps/3733Mbps Co-load,
'L' = 4266Mbps,
'H(by external PU)' = 3733Mbps



Note 13-6: "AUD_DAT_MISO1" is trapping pin to select VEMC Voltage.

AUD_DAT_MISO1	VEMC Voltage
L (Default)	VEMC=3.0V
H (by external PU)	VEMC=2.5V

Title	013_BB_1_ADC/CLK	REV: V10
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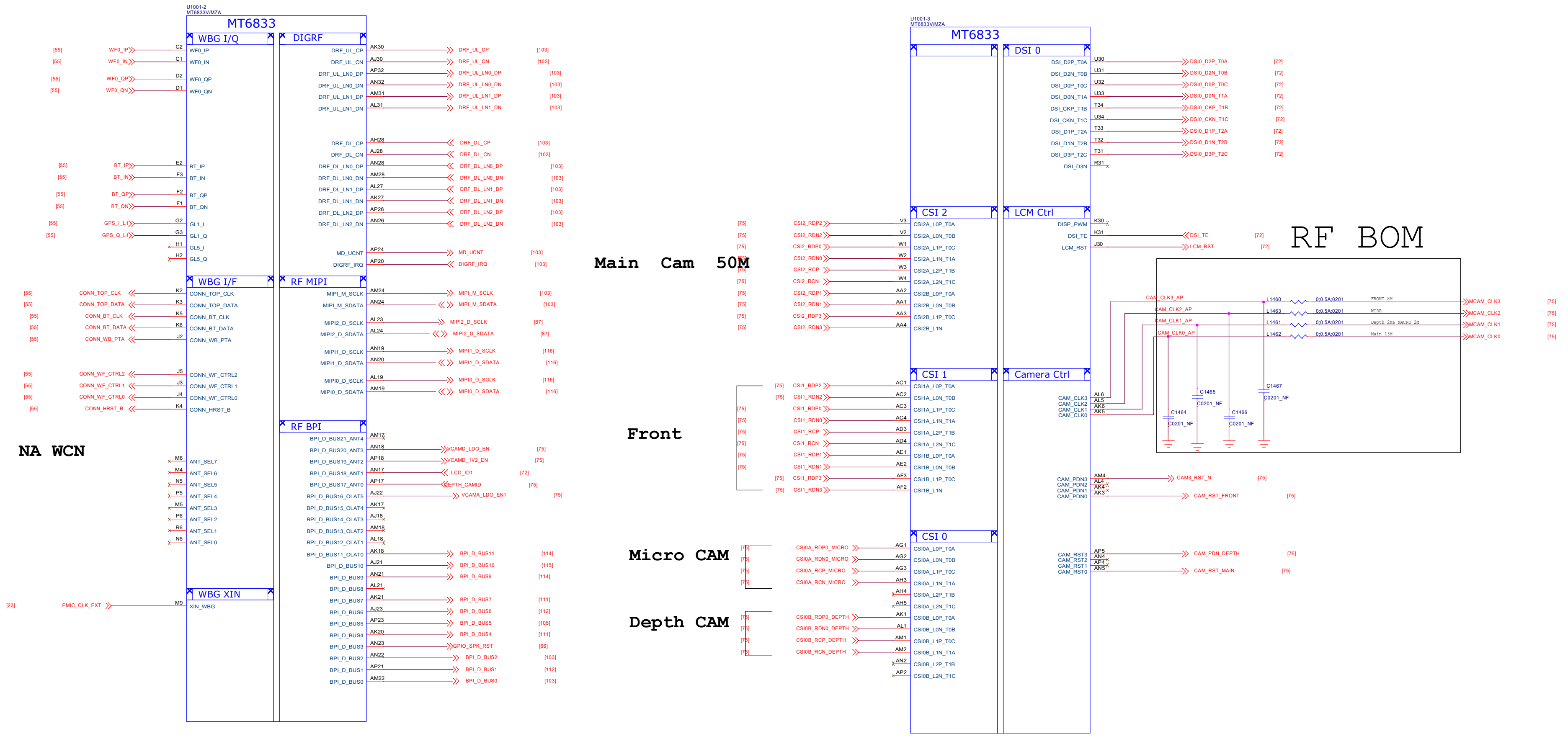
14_BB_2



Title	014_BB_2_MIPI	REV: V10
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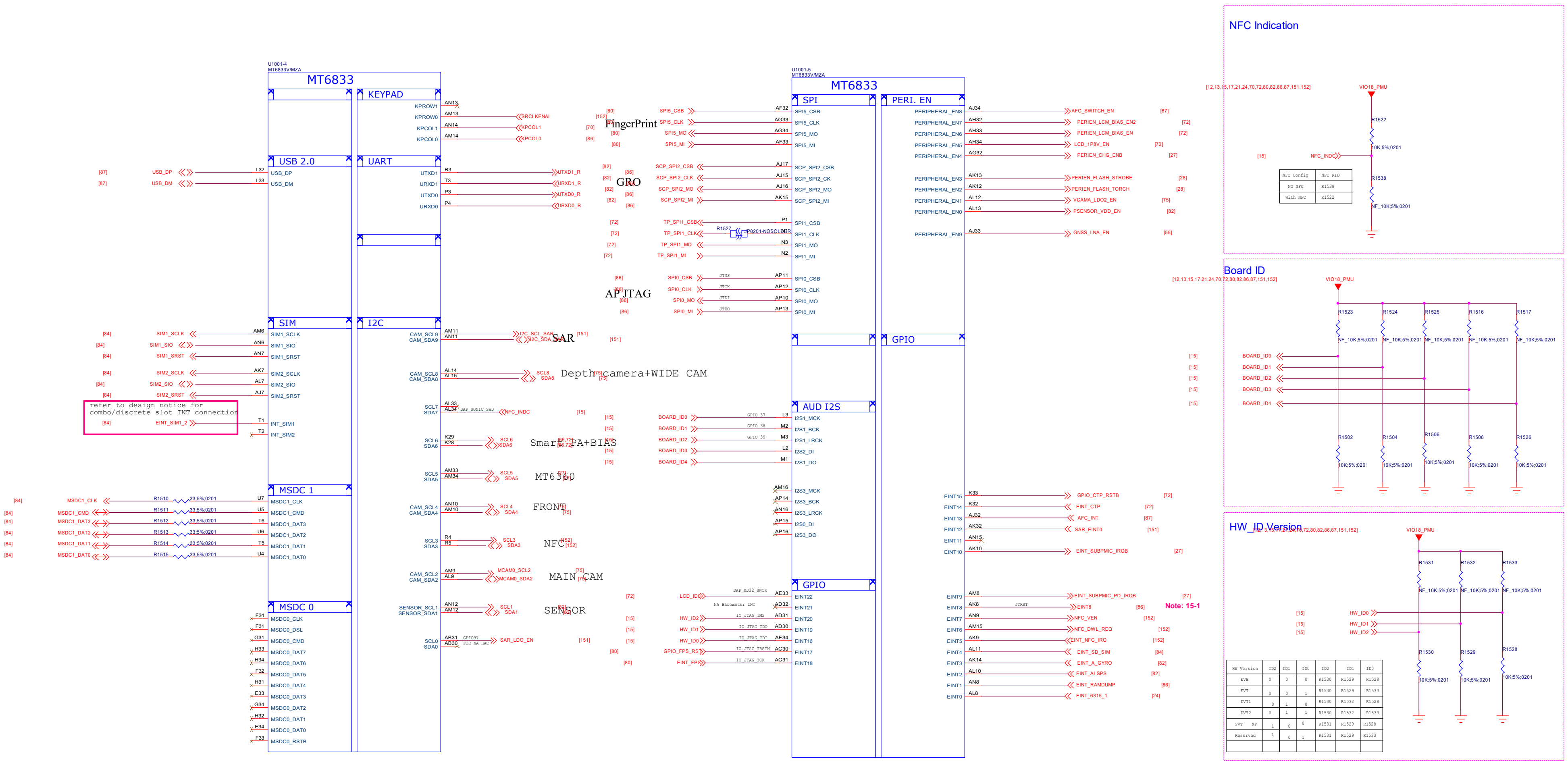
close the SoC

15_BB_3



Title	015_BB_3_MISC	REV: V10
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DEPARTMENT: BB		DESIGNER: liufenglei
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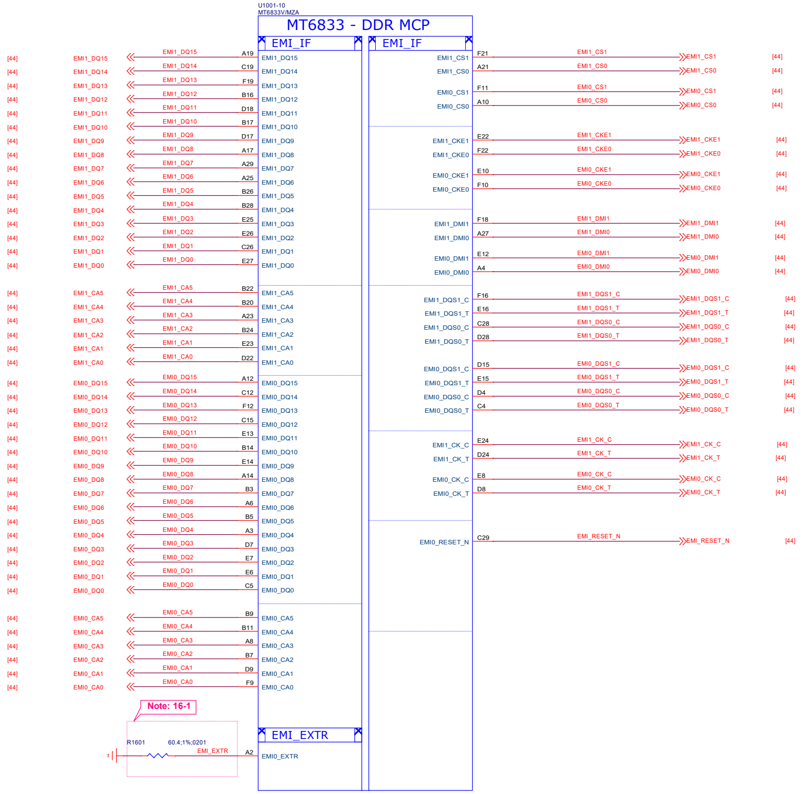
16_BB_4



Schematic design notice of "16_BB_4_Interface" page:

Note 16-1: R4001 please select 60.4 ohm (1%) resistor

Title	016_BB_4_EMI	REV:	V10
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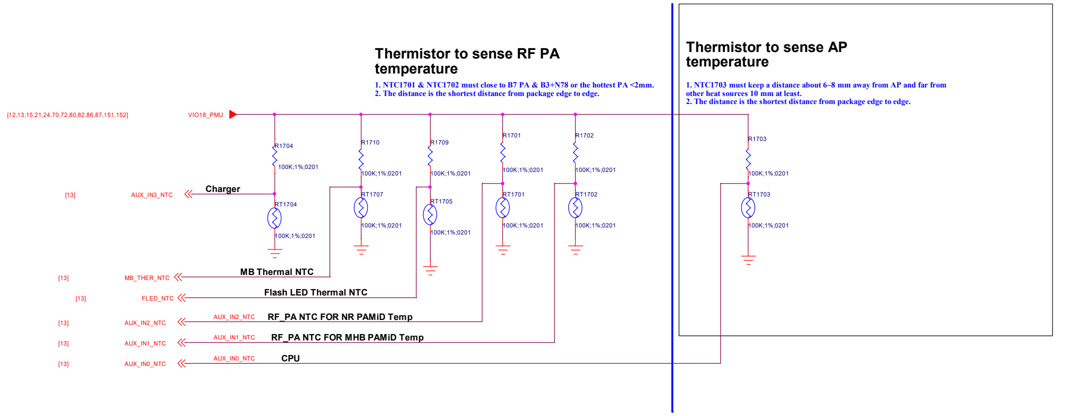
17_BB_AUXADC_Thermal



Title	017_BB_AUXADC_Thermal	REV: V10	
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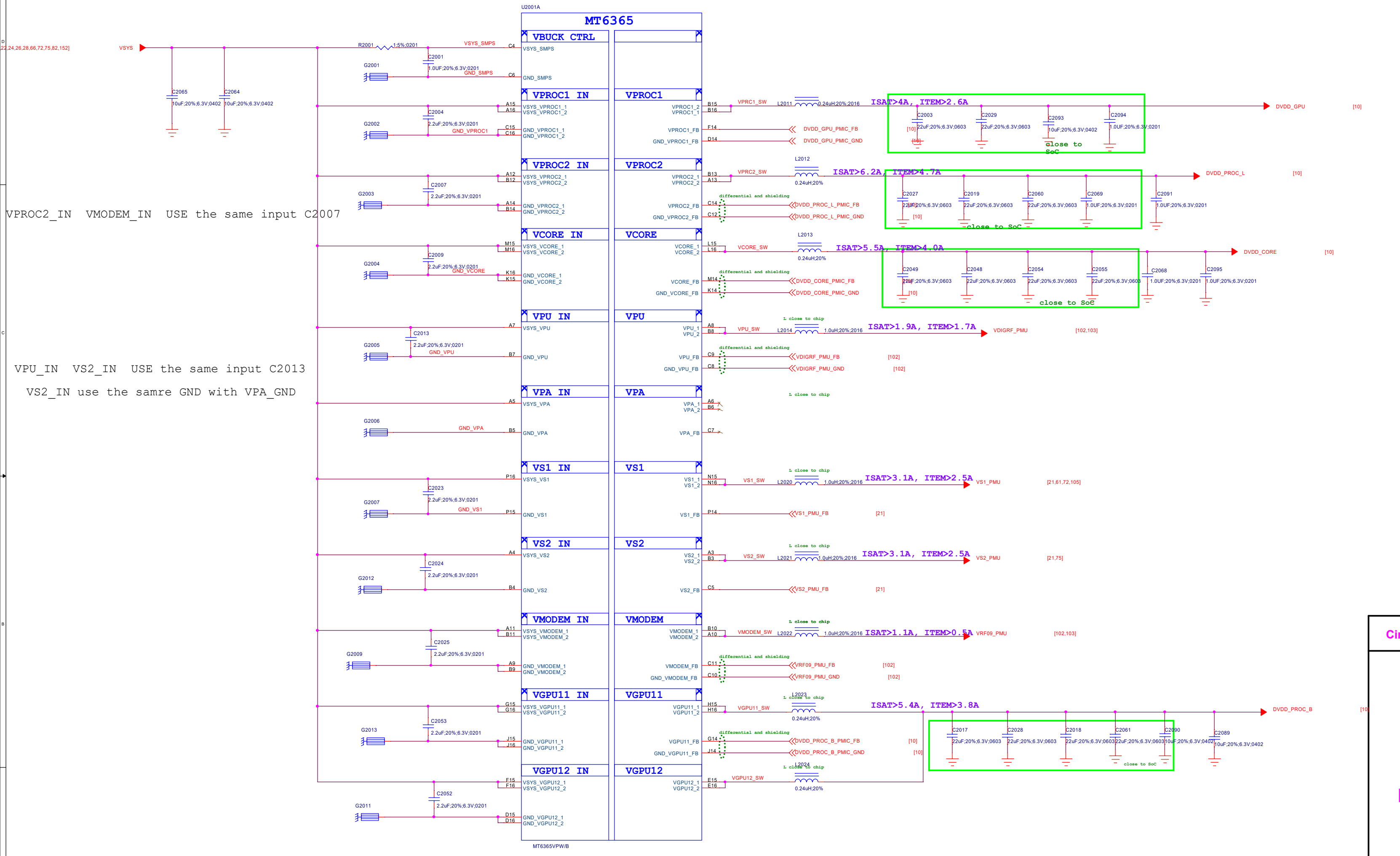
20_POWER_MT6365-Buck



VPROC2_IN VMODEM_IN USE the same input C2007

VPU_IN VS2_IN USE the same input C2013

VS2_IN use the samre GND with VPA_GND

Circuit Type	BUCK Name (Application name)	Output Voltage Range(V)	BOOT Defalut	Iout max(mA)
Buck	VPROC1 (DVDD_GPU)	0.40~1.19 (6.25mV/step)	ON(0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40~1.19 (6.25mV/step)	ON(0.75)	4800
	VGP11+12 (DVDD_PROC_B)	0.40~1.19 (6.25mV/step)	ON(0.75)	4800*2
	VCORE (DVDD_CORE)	0.40~1.3 (6.25mV/step)	ON(0.75)	4800
	VMODEM （VRF09_PMU）	0.50~1.10 (6.25mV/step)	OFF	4800
	VPU (VDIGRF_PMU)	0.40~1.19 (12.5mV/step)	ON(0.7)	2400
	VS1 (VS1_PMU)	1.86~2.20 (12.5mV/step)	ON(2.0)	2200
	VS2 (VS2_PMU)	1.20~1.50 (12.5mV/step)	ON(1.35)	2500
	VPA (VPA_PMU)	0.50~3.40 (50mV/step)	OFF	1000

Title	020_POWER_MT6365_Buck	REV:	V10
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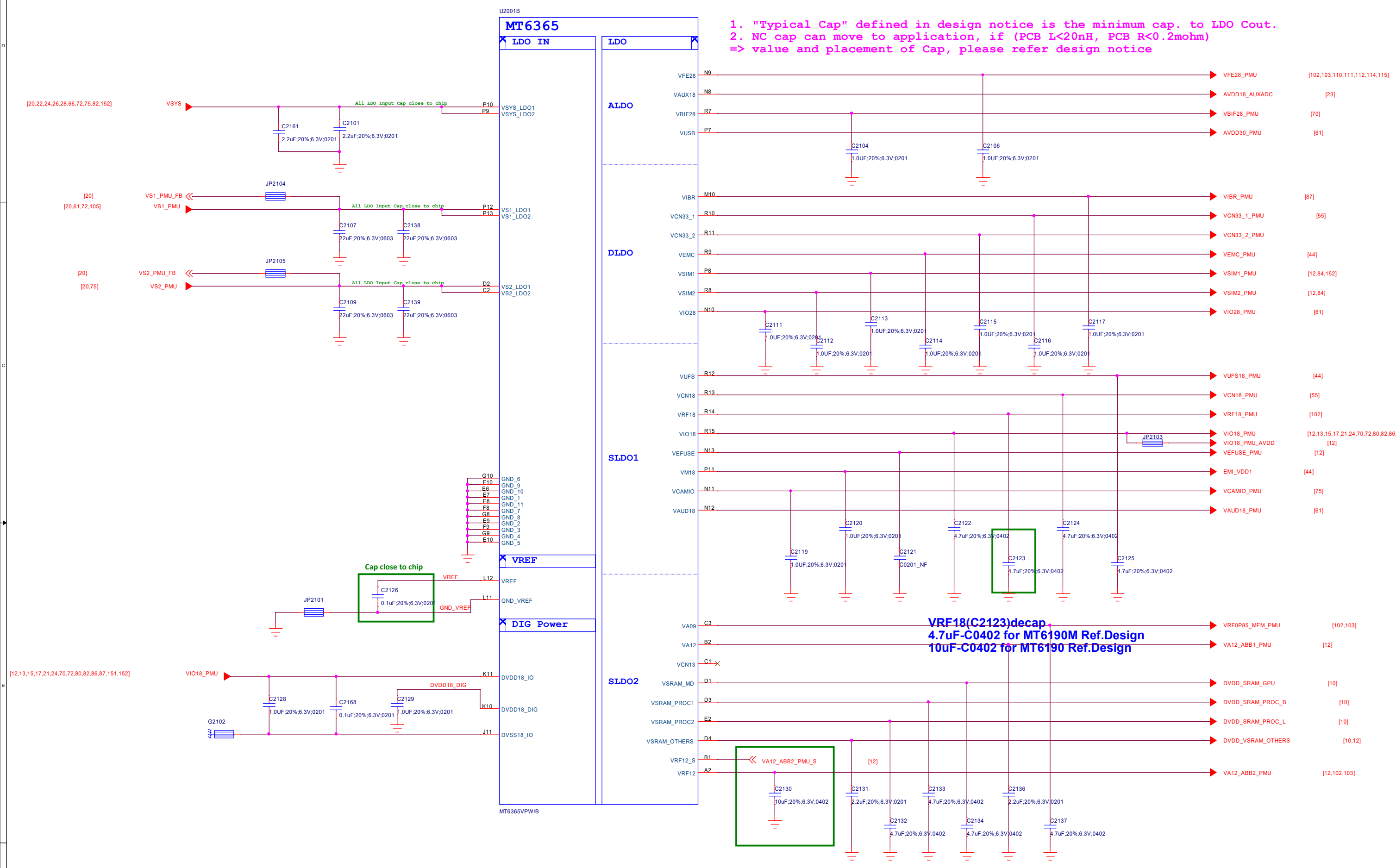
21_POWER_MT6365-LDO

1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2mohm)
=> value and placement of Cap, please refer design notice



LDO Name	Output Voltage(V)	BOOT Defalut	Iout max(mA)	Expected use
VEF28	2.80	OFF(2.8)	200	RFFE
VUSB	3.07	ON(3.07)	200	USB
VAUX18	1.84	ON(1.84)	50	AUXADC
VXO22	2.24	ON(2.24)	25	DCXO
VRFCK	1.4@boot 1.6@work	ON(1.4)	10	DCXO
VBIF28	2.80	OFF(2.8)	50	Battery Interface
VRTC	2.80	ON(2.8)	2	RTC
DVDD18_DIG	1.80	ON(1.8)	10	PMIC Digital
VSIM1	1.7/1.8/1.86/2.76/3.0/3.1	OFF(1.86)	200	SIM
VSIM2	1.7/1.8/1.86/2.76/3.0/3.1	OFF(1.86)	200	SIM
VCN33_1	3.3/3.4/3.5/3.6	OFF(3.3)	800	Connectivity
VCN33_2	3.3/3.4/3.5/3.6	OFF(3.3)	800	Connectivity
VEMC	2.55/2.9/3.0/3.3	ON(3.0)	800	eMMC/UFS
VIO28	2.8/2.9/3.0/3.1/3.2/3.3	OFF(2.8)	200	IO&Sensor
VIBR	2.7/2.8/3/3.3	OFF(2.8)	200	Vibrator
VEFUSE	1.80	OFF(1.8)	300	EFUSE
VAUD18	1.80	ON(1.8)	300	Audio
VCAMIO	1.80	OFF(1.8)	300	Camera IO
VM18	1.80	ON(1.8)	300	DRAM
VRF18	1.80	OFF(1.8)	450	RF
VIO18	1.80	ON(1.8)	600	IO&Sensor
VCN18	1.80	OFF(1.8)	1200	Connectivity
VUFS	1.86	ON(1.86)	1200	eMMC/UFS
VBBCK	1.24	ON(1.24)	10	DCXO
VA09	0.85	ON(0.85)	300	DIGRF SRAM (VRF0P85_MEM_PMU)
VA12	1.2	ON(1.2)	300	AP Analog Module
VCN13	1.3	OFF(1.3)	350	Connectivity
VSRAM_PROC1	0.60~1.2 （6.25mV/step）	ON(0.85)	600	CPUB SRAM
VSRAM_PROC2	0.60~1.2 （6.25mV/step）	ON(0.85)	600	CPUL SRAM
VSRAM_OTHERS	0.60~1.2 （6.25mV/step）	ON(0.75)	600	OTHERS SRAM
VSRAM_MD	0.60~1.2 （6.25mV/step）	ON(0.85)	600	GPU SRAM
VRF12	1.2	ON(1.2)	800	AP Analog Module

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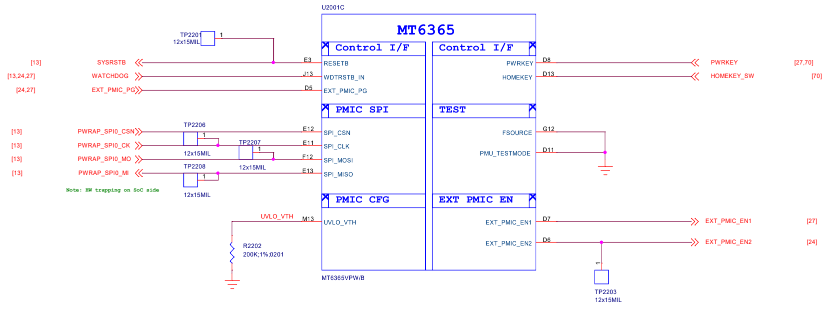
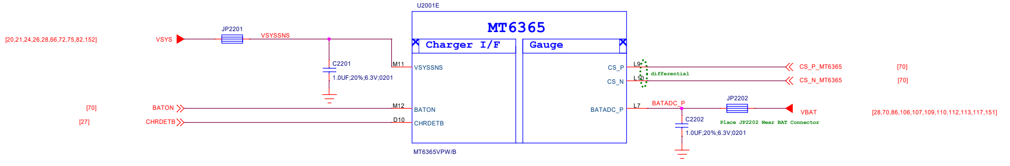
22_POWER_MT6365-General





Title	022_POWER_MT6365_General	REV:	V10
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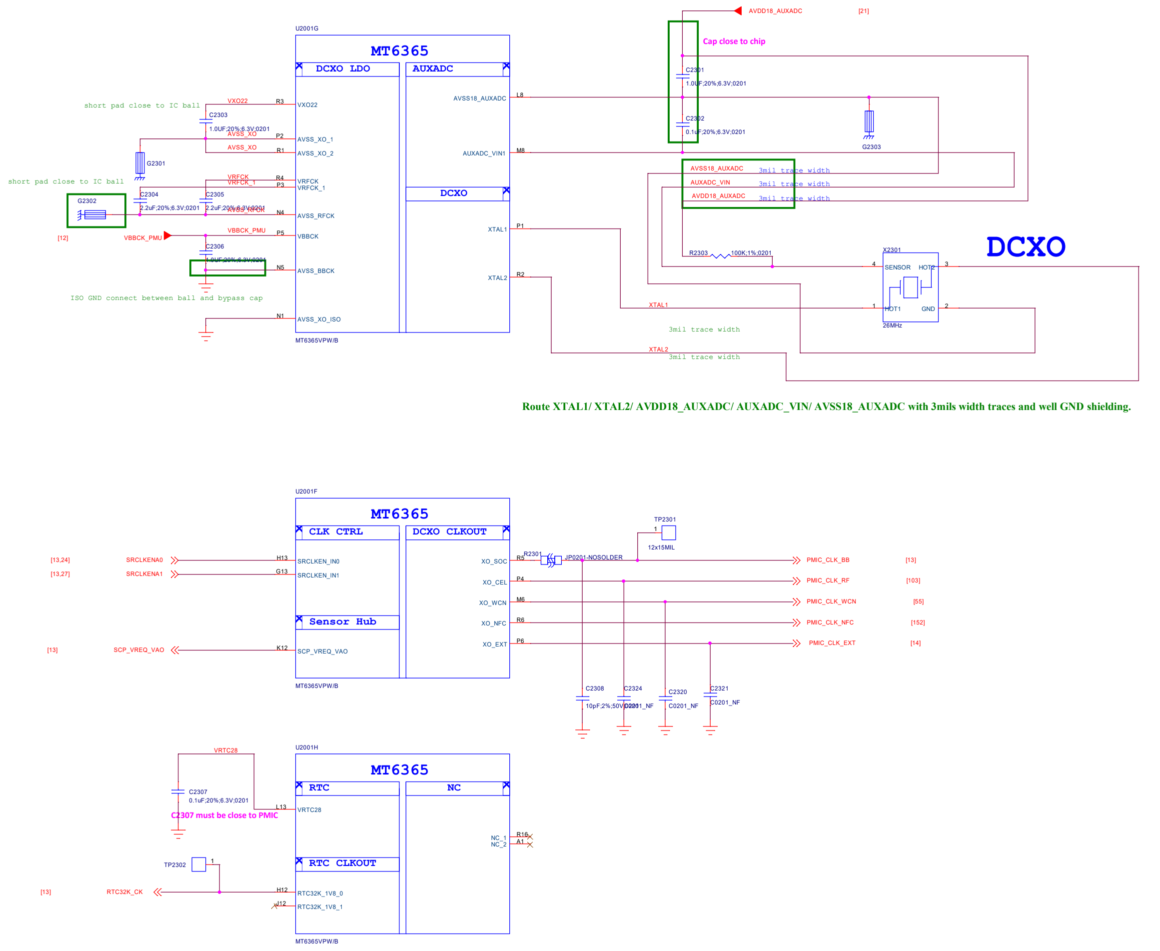
23_POWER_MT6365_Clock



Title	023_POWER_MT6365_Clock	REV:	V10
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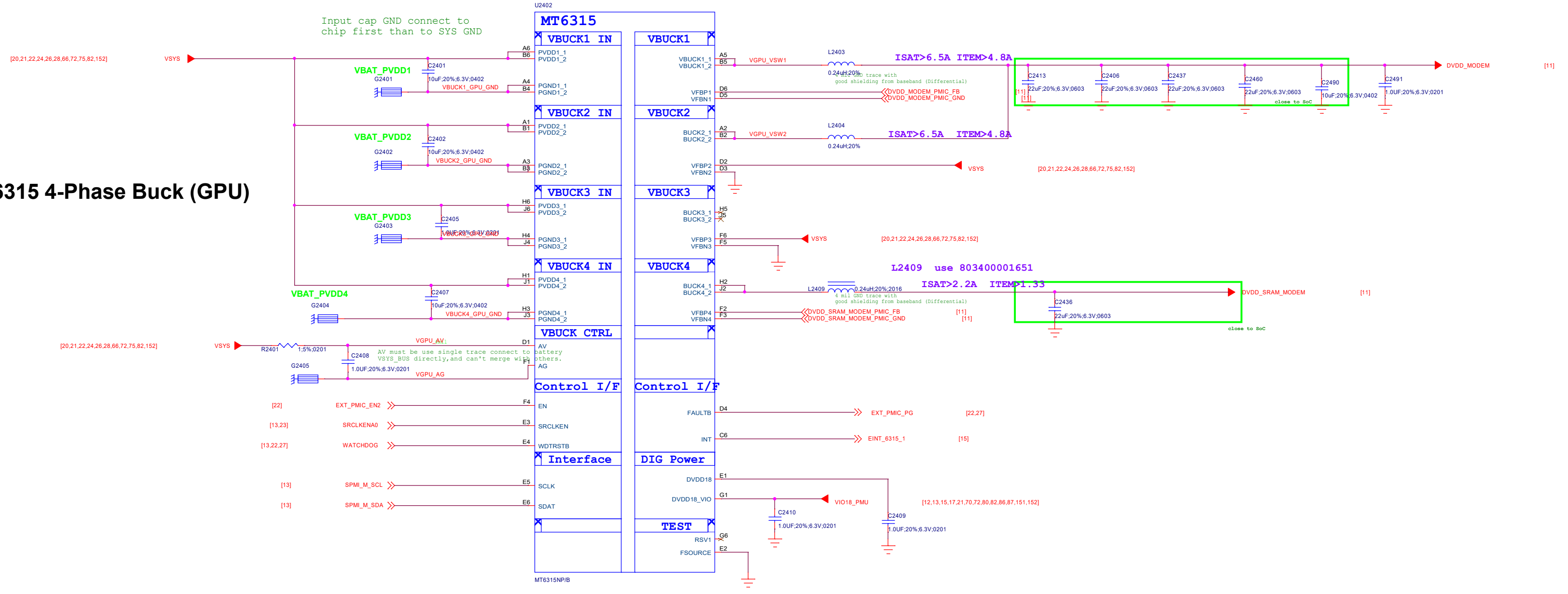
MT6315 4-Phase Buck (GPU)



Title	024_POWER_MT6315_BUCK		REV: V10
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Title	025_Reserved	REV: V10
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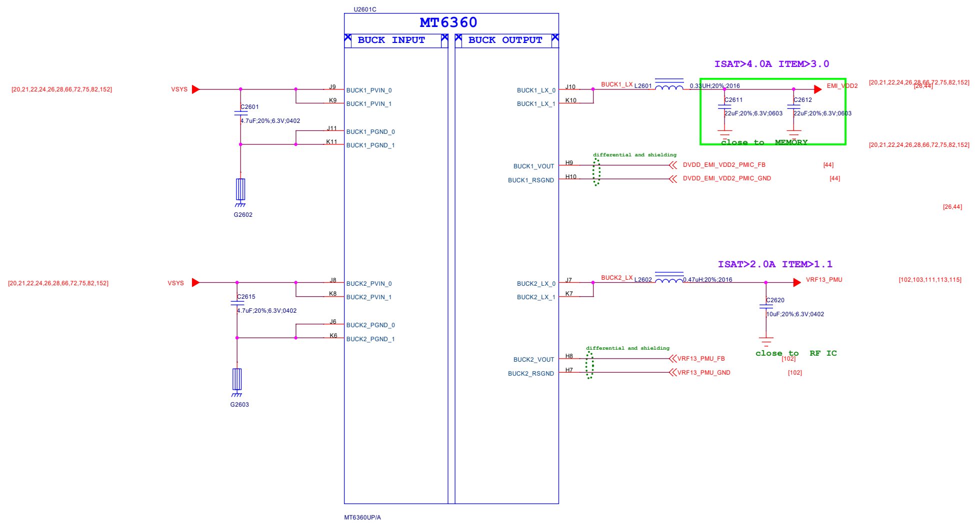
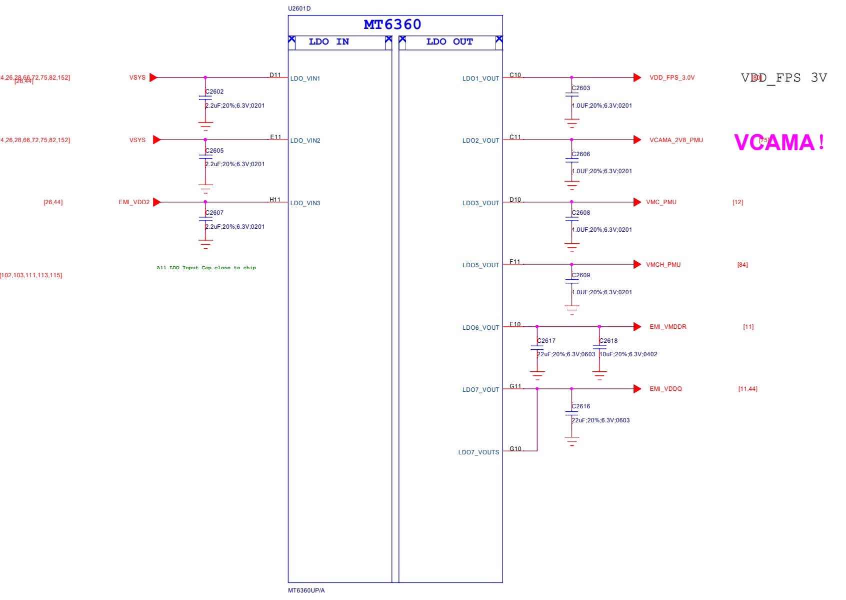
26_POWER_MT6360_BUCK_LDO





Circuit Type	Name	Output Voltage Range(V)	BOOT Defalut	Iout max(mA)	Expected use
Buck	BUCK1	0.3~1.3 (5mV/step)	ON(1.125)	3000	VDRAM1
	BUCK2	0.3~1.3 (5mV/step)	ON(0.75)	3000	VSRAM_CORE+ADSP
LDO	LDO1	1.8/2/2.1/2.5/2.7/ 2.8/2.9/3/3.1/3.3	OFF(1.8)	150	Fingerprint
	LDO2	1.8/2/2.1/2.5/2.7/ 2.8/2.9/3/3.1/3.3	OFF(1.8)	200	CAM AVDD
	LDO3	1.8/2.9/3/3.3	OFF(3.0)	200	SD Card
	LDO5	2.9/3/3.3	OFF(3.0)	800	SD Card
	LDO6	0.75	ON(0.75)	300	EMI_VMDDR
	LDO7	0.6	ON(0.6)	600	EMI_VDDQ

Title: 026_POWER_MT6360_LDO_BUCK
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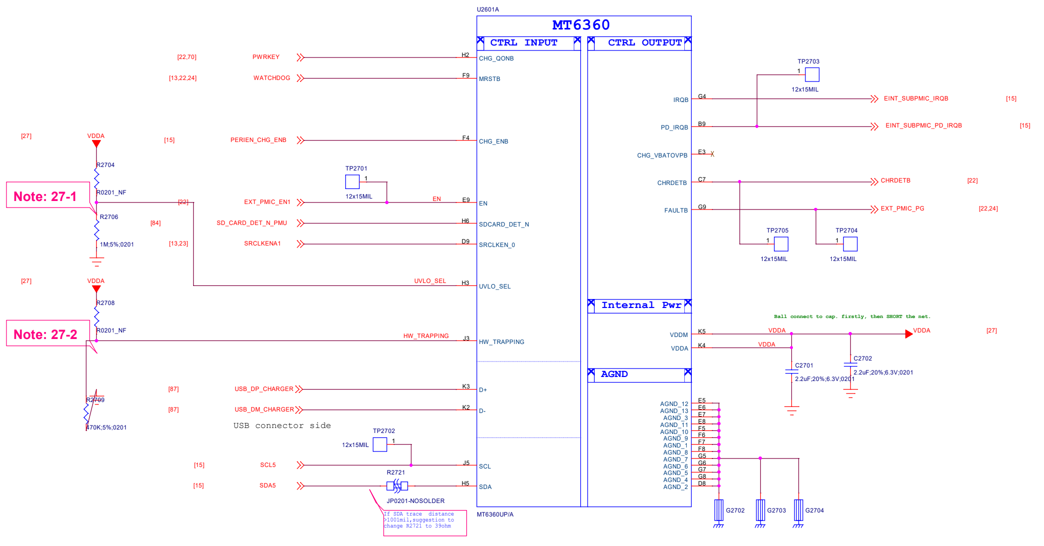
27_POWER_MT6360_General



Note 27-1: UVLO_SEL pin is define difference UVLO power configuration.The power configuration is used to Ruvlosel (R2706) to select.

Note 27-2: HW_TRAPPING is define difference power configuration.The power configuration is used to Rhw_trapping (R2709&R2704) to select.

REG_PMU0x04[7:5] : HW_SYSUV_SEL[2:0]	RUVLO_SEL Range(Ω)		
	Min	Typ	Max
000（2.8V）	Short to VDDA		
001（2.9V）	936K	1.8M	---
010（3.0V）	234K	430K	663K
011（3.1V）	58K	100K	165K
100（3.2V）	14K	28K	41K
101~111（3.3V）	---	7.5K	10K

REG_PMU0x99[2:0] : HW_TRAPPING[2:0]	Rhw_trapping Range(Ω)			BUCK1	BUCK2	LDO6	LDO7
	Min	Typ	Max				
000	Short to VDDA			0.725V	1.125V	0.75V	0.6V
001	936K	1.8M	---	0.725V	1.125V	X	1.8V
010 (Default)	234K	430K	663K	1.125V	0.725V	0.75V	0.6V
011	58K	100K	165K	1.125V	0.725V	X	1.8V
100	14K	28K	41K	0.725V	1.225V	X	X
101	3.8K	7.5K	10K	1.225V	0.725V	X	X
110	0.96K	1.6K	2.4K	0.725V	1.125V	X	X
111	---	422	603	0.725V	0.725V	X	X

Title	027_POWER_MT6360_General
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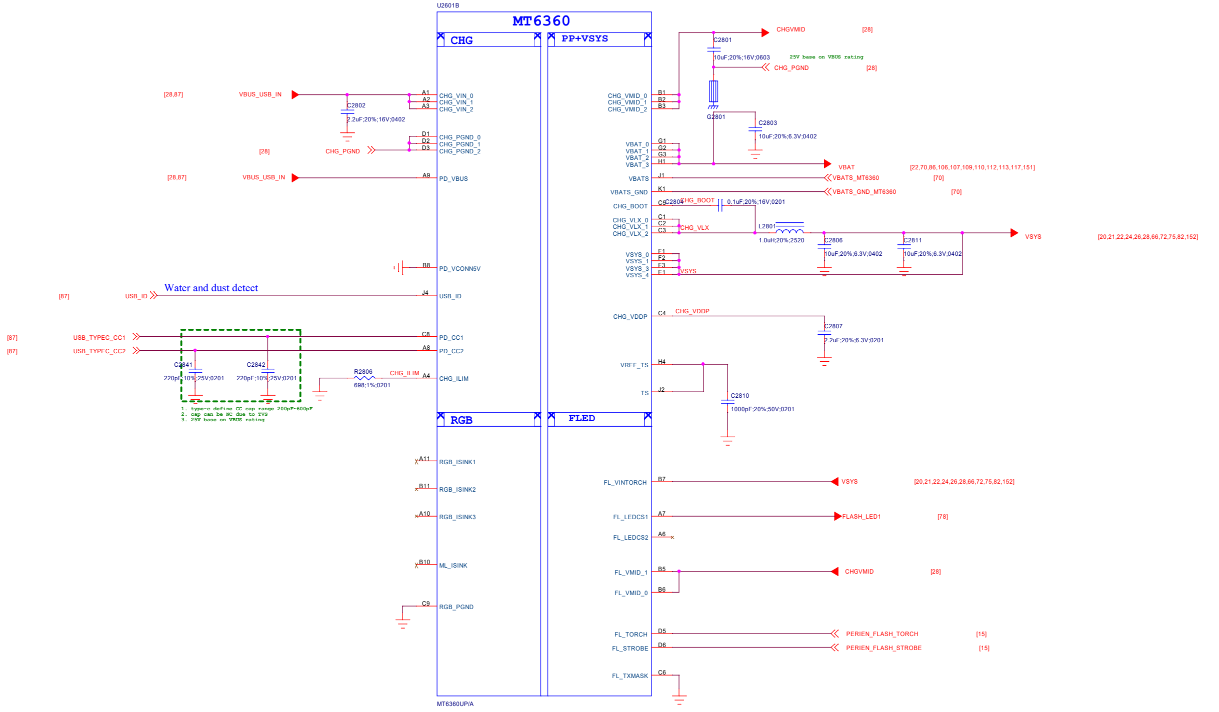
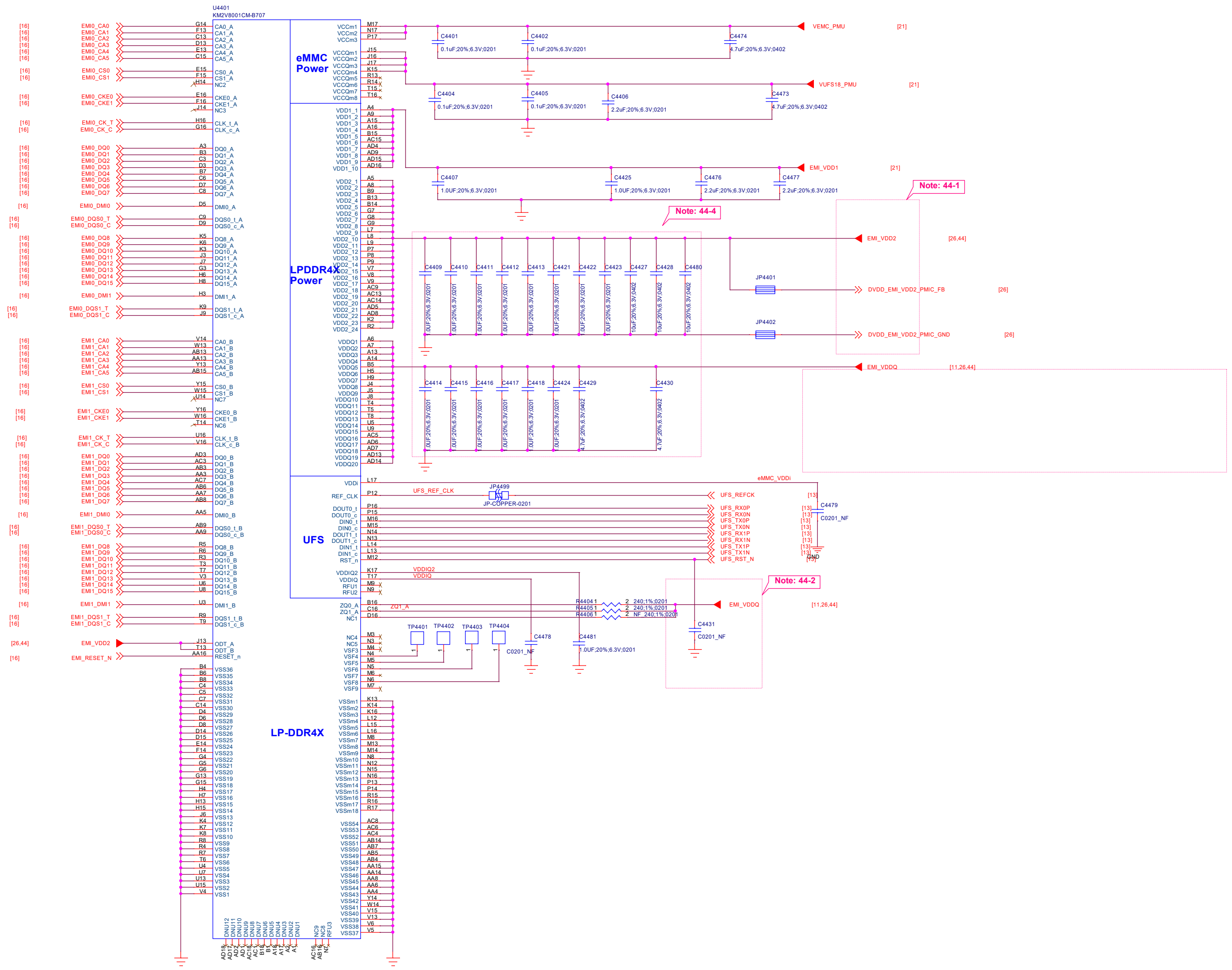
28_POWER_MT6360_Charger



Title	028_POWER_MT6360_Charger		REV: V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446		Size D
DEPARTMENT:	BB	DESIGNER:	liufenglei
Date:	Friday, September 16, 2022	Sheet	28 of 152

44_Memory_eMMC_LPDDR4X



Schematic design notice of "44_Memory_UFS_LPDDR4X" page.

Note 44-1: Please refer to power supply related page select LDO7_VOUT / BUCK1_LX output voltage properly for LPDDR4X

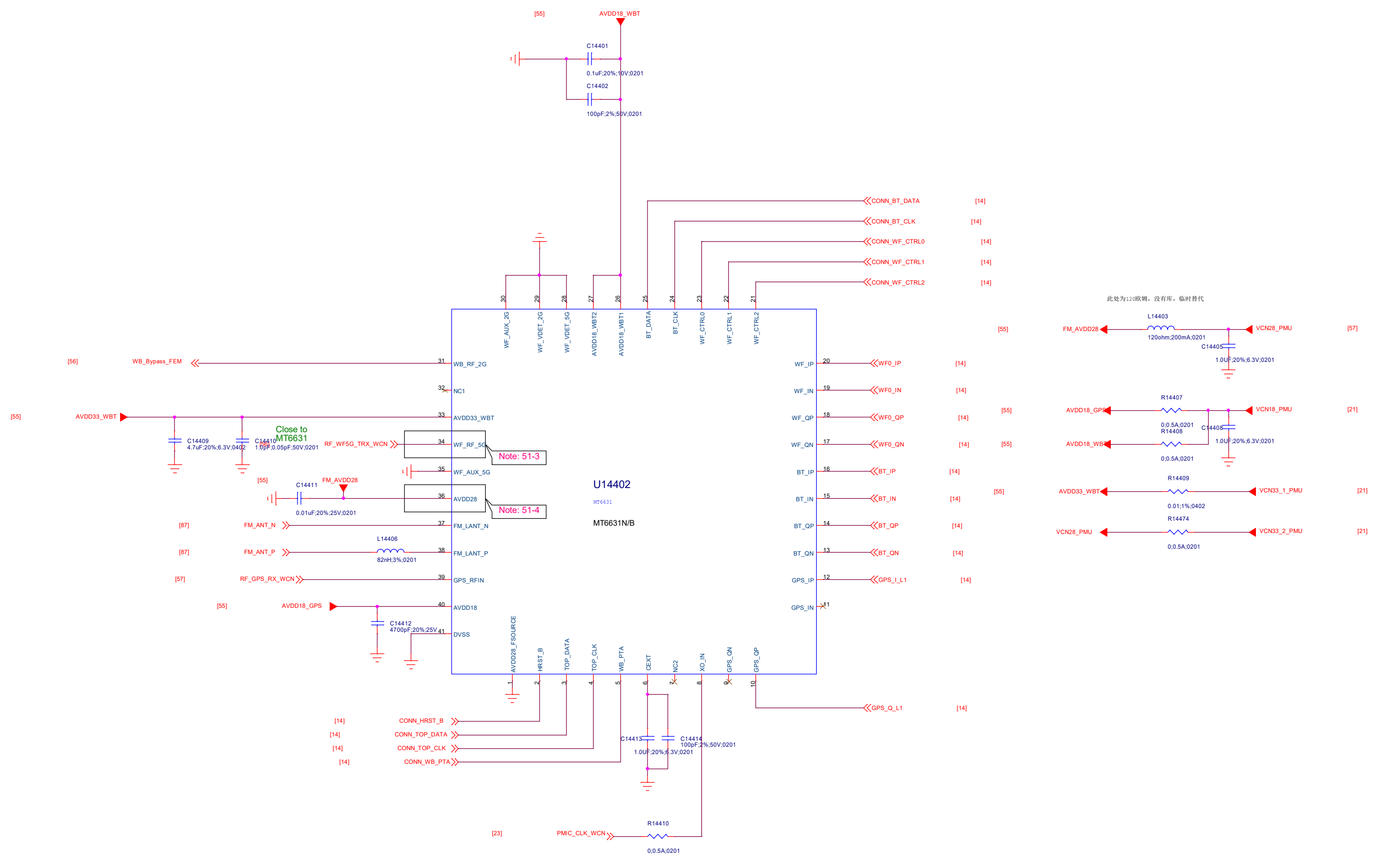
Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to uMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of UFS.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

Title	044_Memory_eMMC_LPDDR4X		V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT: BB		DESIGNER:	liufenglei
Date:	Friday, September 16, 2022	Sheet 44	of 152

WINGTECH

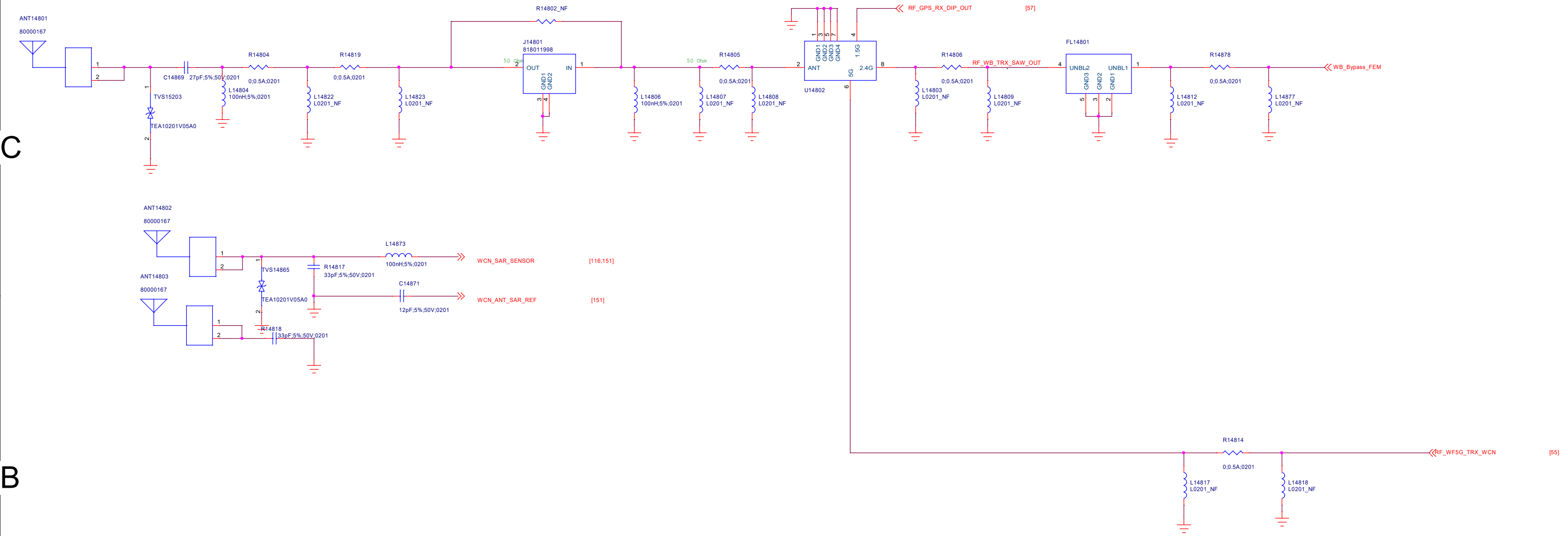


Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"

Note 51-1:	For R5015 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6762 Baseband design notice for VCN33 LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support
Note 51-5:	If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

Title	144_WCN_MT6631	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	WINGTECH-SZ	DESIGNER:	PENGLIZHONG
Date:	Friday, September 16, 2022	Sheet	55 of 59

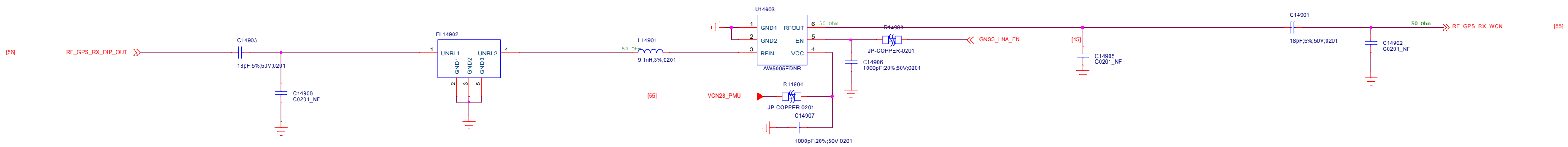
2.4G Wi-Fi BT/5G WIFI



Title	148_WIFI_FEM	REV: V10
DOCUMENT NO.:	Design Name = 96901_1_11_M12_20220916_1446	Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: PENGLIZHONG
Date:	Friday, September 16, 2022	Sheet 56 of 59

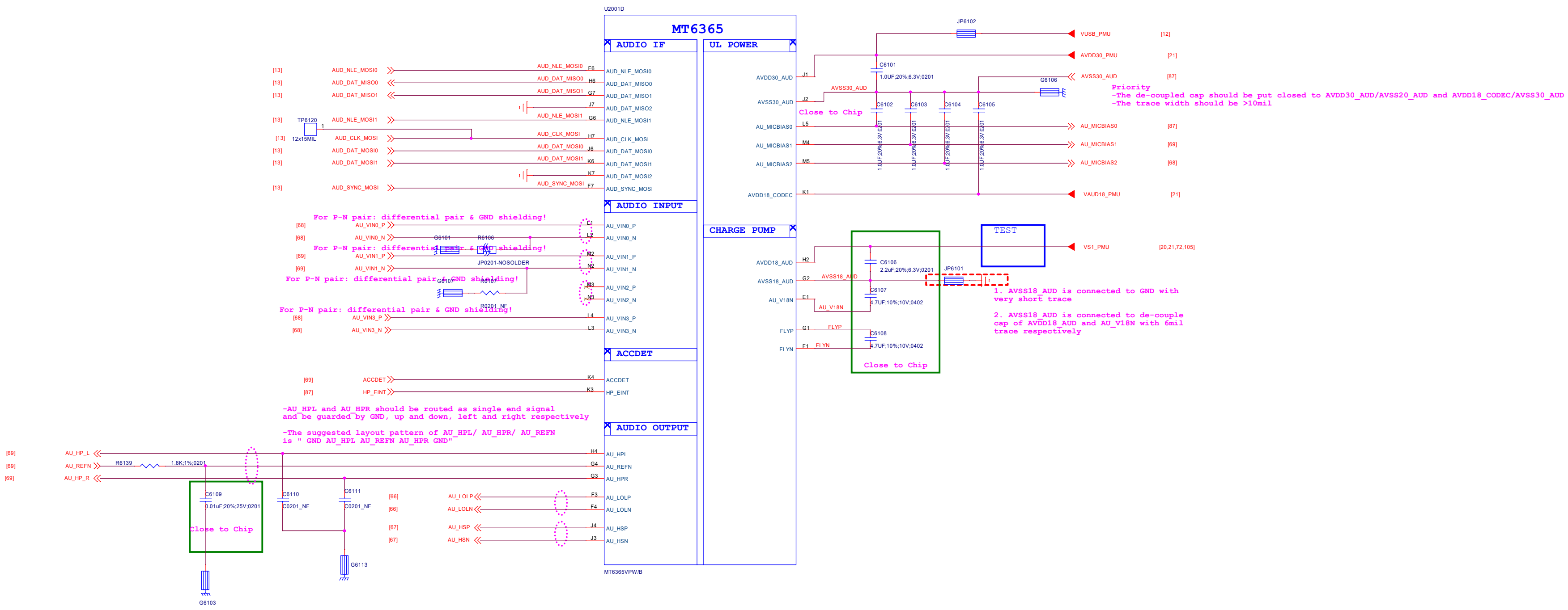
Close to MT6331

GPS xLNA



Title	149_GPS	REV:	V10
DOCUMENT NO.:	Design Name = 96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	PENGLIZHONG
Date:	Friday, September 16, 2022	Sheet 57	of 59

61_POWER_MT6365_AUDIO



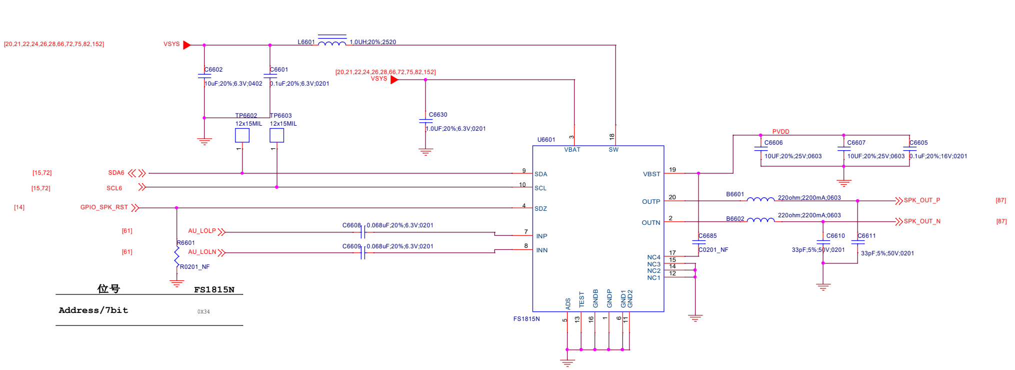
Title	061_PERI_6365_AUDIO_PMIC-IF	Rev	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	BB	DESIGNER:	liufenglei
Date:	Friday, September 16, 2022	Sheet	61 of 152

WINGTECH

066_SPK

FOR NA

FOR Global



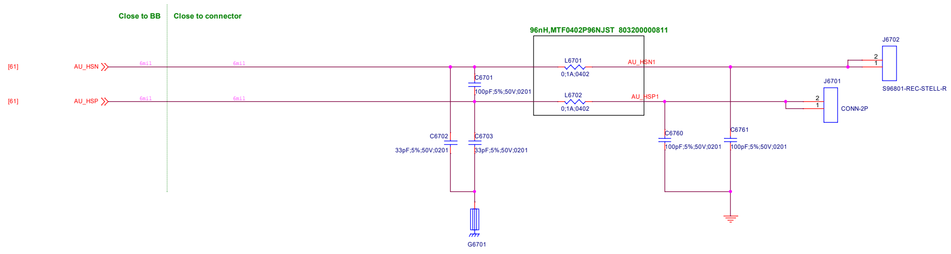
位号	FS1815N
Address/7bit	0X34

Title	066_SPK	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT: BB		DESIGNER:	Ning Xinwu
Date:	Friday, September 16, 2022	Sheet 66	of 57

WINGTECH

067_Receiver

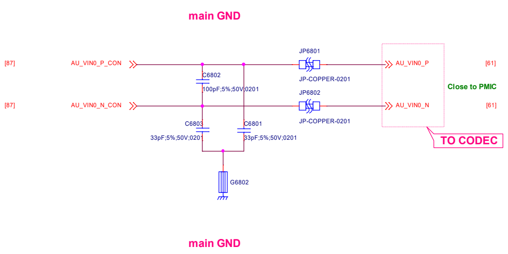
Receiver



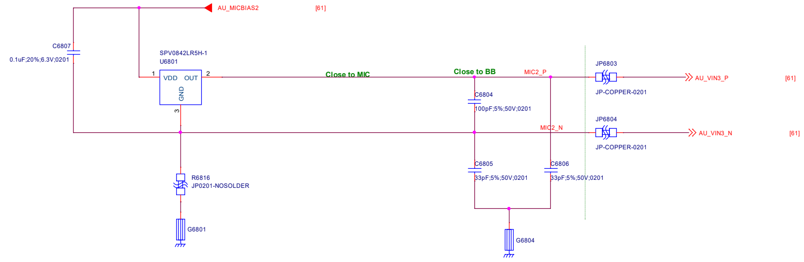
Title	067_Receiver	REV: V10	
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size D	
DEPARTMENT:	BB	DESIGNER:	liufenglei
Date:	Friday, September 16, 2022	Sheet 67	of 68

068_MIC

MAIN MIC



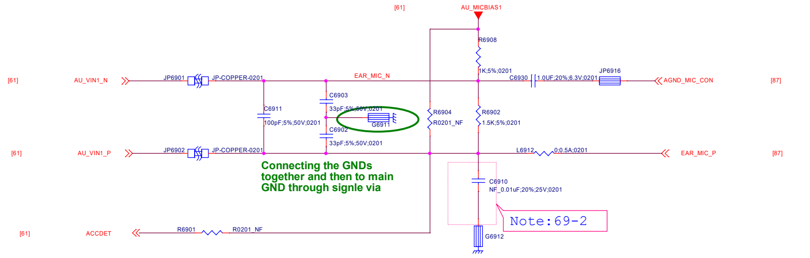
second Microphone



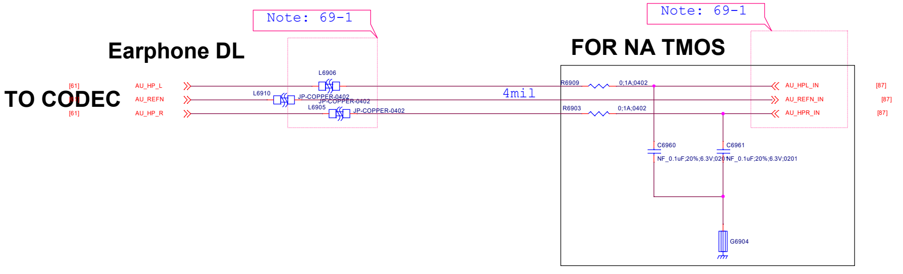
Title	068_MIC	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	BB	DESIGNER:	liufenglei
WINGTECH			
Date:	Friday, September 16, 2022	Sheet	68 of 152

069_Earphone

Earphone Microphone



L6905,L6906,L6910 上磁珠803340000181 debug



Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 69-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 69-2: Reserved Cap C6910 for CS/RS test, please double check multi-key function when used

Note 4-3:

	Earphone Jack @ Main board	Earphone Jack @ Sub board
R413	0 ohm	100nF

Note 4-4:
1. Need do SW configuration for DCC mode
2. Analog input扭 impedance is as below, please customer check the detail requirement with mic vendor if the mic used has special requirement on the input impedance of analog input

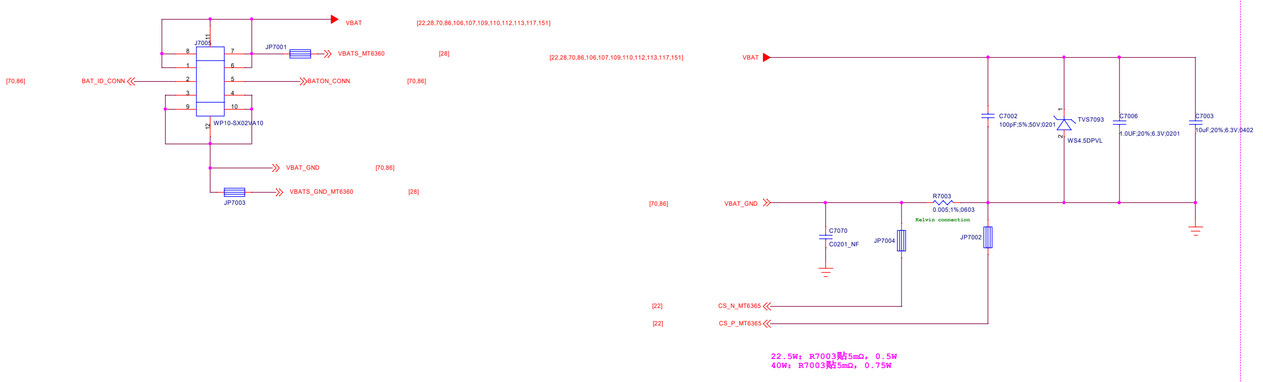
MT6357	UL Mode	MIN	TYP	MAX	Unit
C	DCC	-	-	11p	F

Note 4-5: Please select R6231 with 0402 size

Title	069_Earphone	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	BB	DESIGNER:	liufenglei
Date:	Friday, September 16, 2022	Sheet	69 of 152

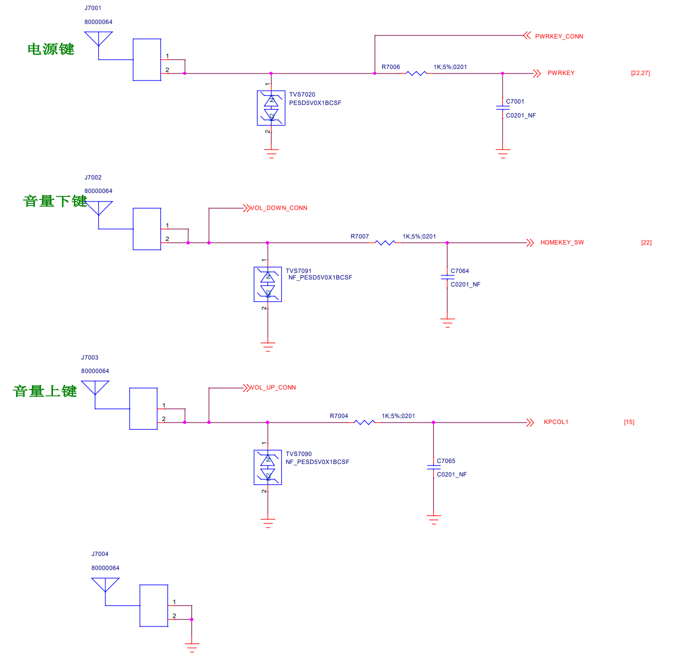
070_Battery/Sidekey

BATTERY CONNECTOR

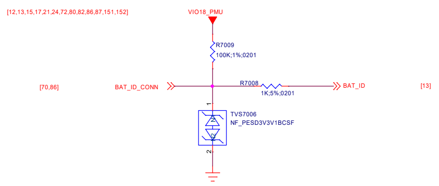


Power Key / Key Pad

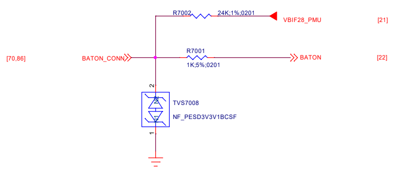
DO NOT put pull-up resistor on PWRKEY



BAT ID



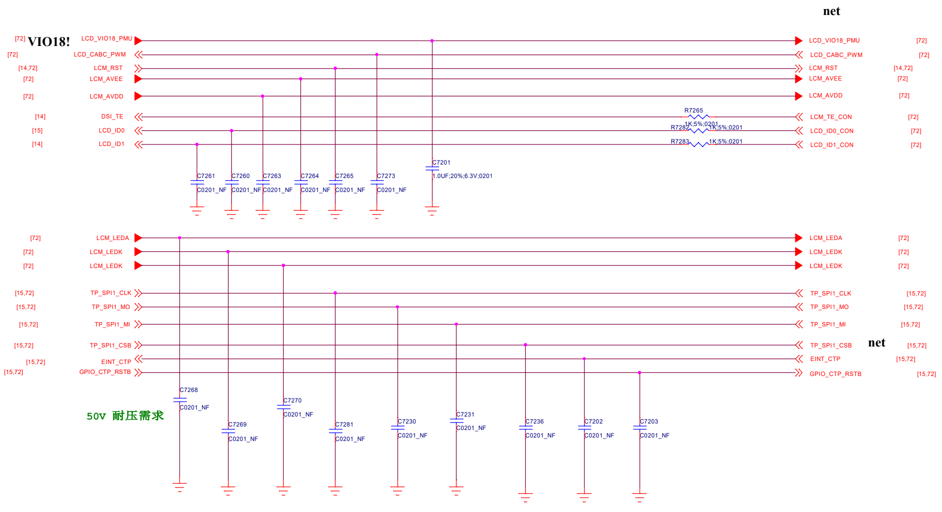
Thermal



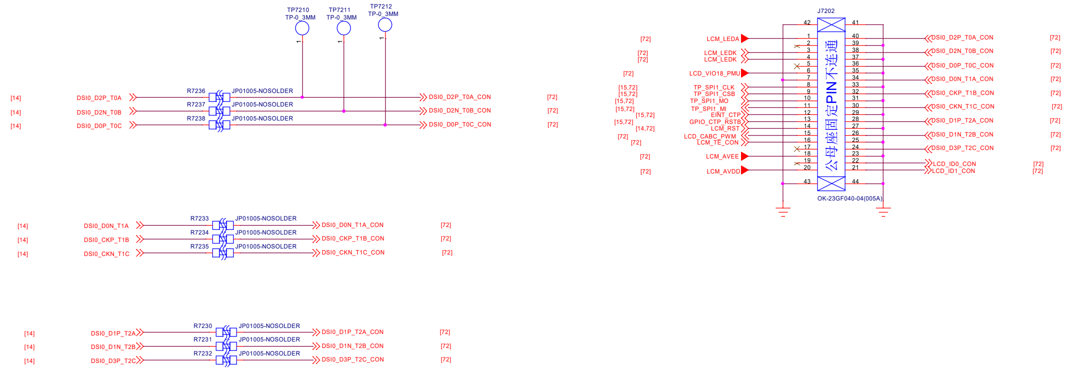
Water_Det_Key

Title	070_Battery/Sidekey	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT: BB		DESIGNER:	liufenglei
WINGTECH			
Date:	Friday, September 16, 2022	Sheet 70	of 152

072_LCD/CTP IF

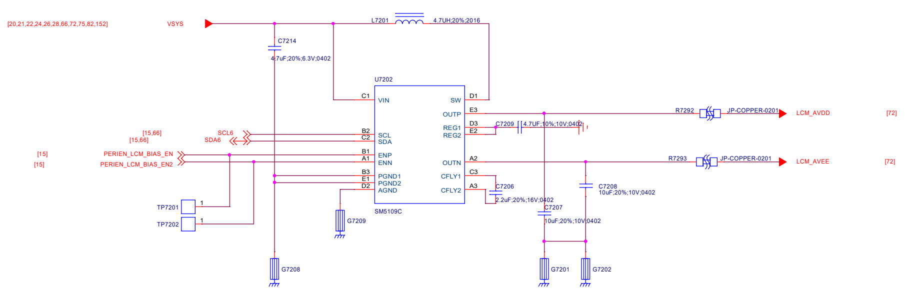


Main LCM



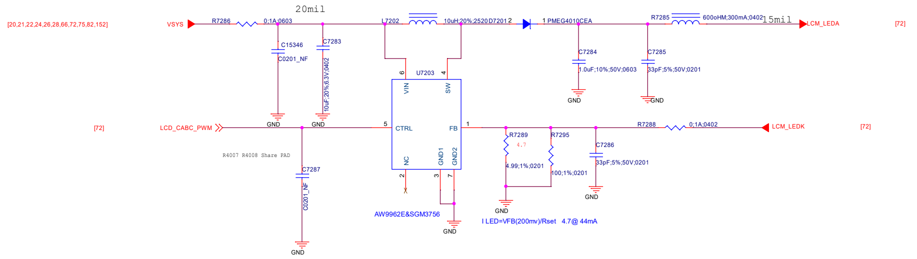
LCM Gate Drive

SM5109/OCP2130 LCM Gate Driver I2C address: 0X3E (Write:0x7C, Read:0x7D)

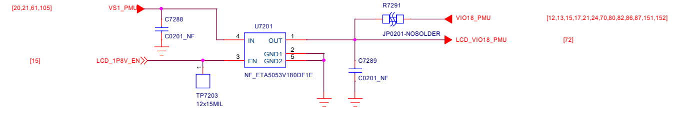


LCD-BACKLIGHT

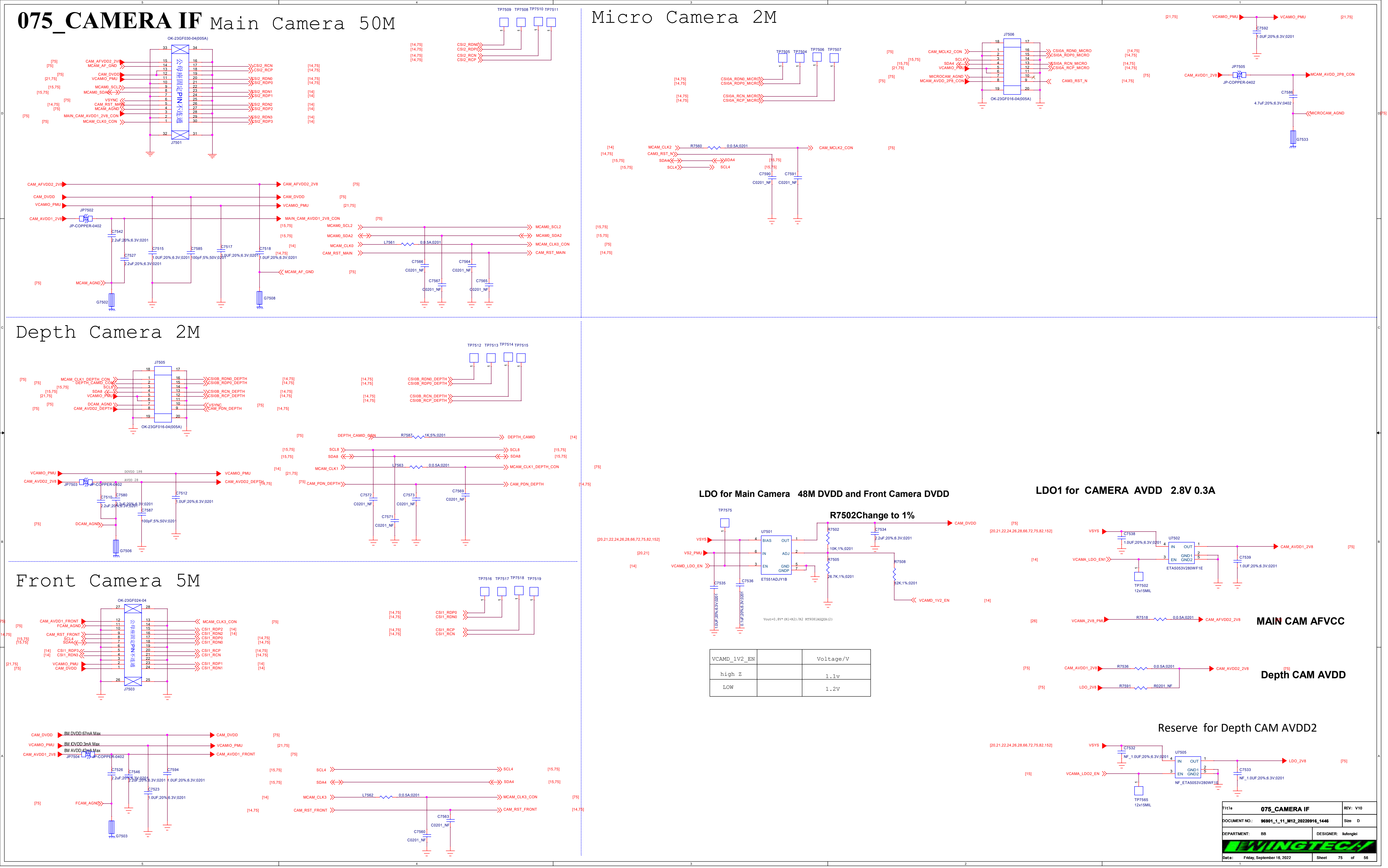
Rating : 50V



Reserve for ESD



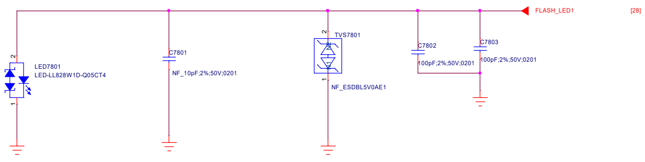
Title	072_LCD/CTP IF	REV: V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size D
DEPARTMENT: BB		DESIGNER: liufenglei
WINGTECH		
Date: Friday, September 16, 2022		Sheet 72 of 38



VCAMD_1V2_EN		Voltage/V
high Z		1.1v
LOW		1.2V

078_Flash/RGB/Vibrator

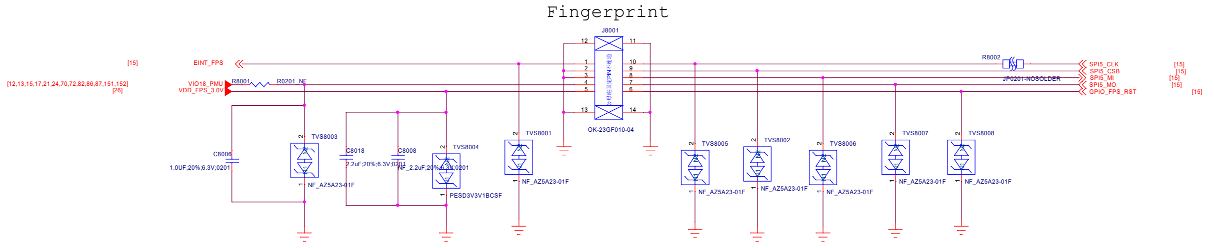
Flash LED



HAPTIC & ERM

Title	078_Flash/Vibrator	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	BB	DESIGNER:	liufenglei
Date:	Friday, September 16, 2022	Sheet	78 of 152

080_FingerPrint

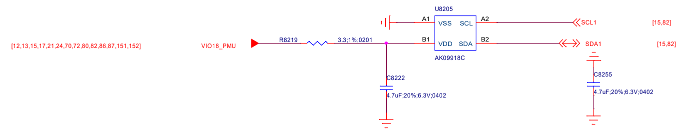


Title	080_FingerPrint	REV: V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size D
DEPARTMENT: BB		DESIGNER: liufenglei
Date:	Friday, September 16, 2022	Sheet 80 of 68

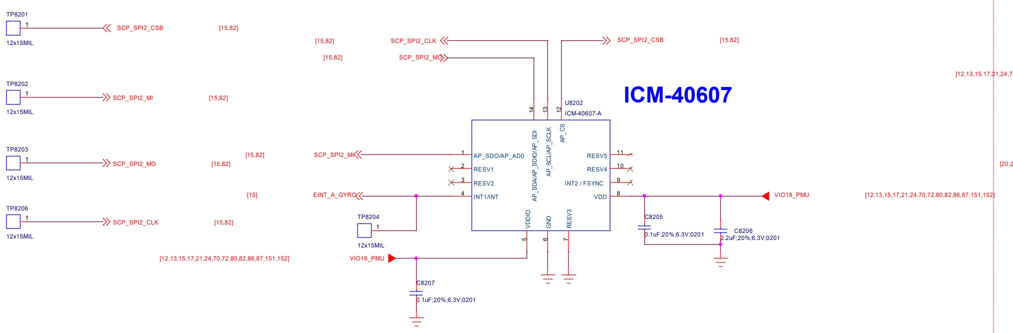
082_SENSORS

Accerometer

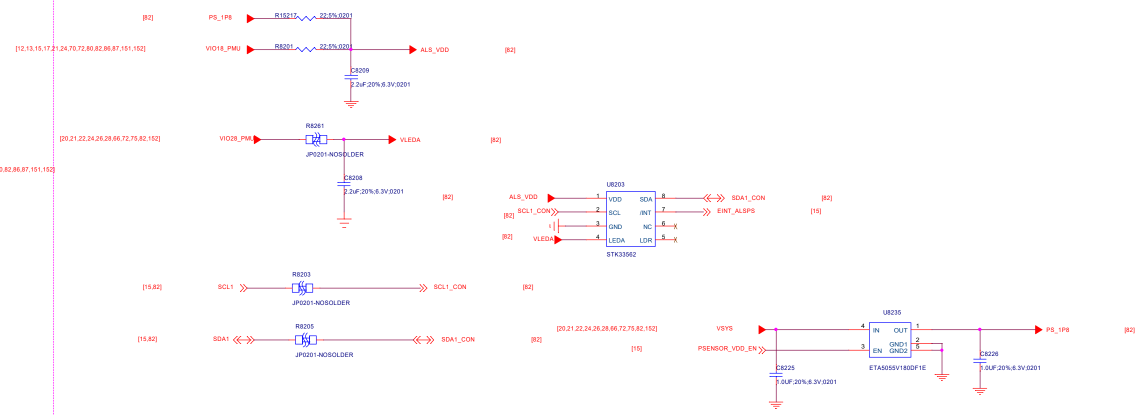
E-compass



Accerometer + Gyro Sensor Optional



ALS+PS TMD27123



Schematic design notice of "82_PERI_SENSORS_MEMs_ALS/PS" page.

Note 82-1: [M sensor] Keep a minimum distance of 15mm from power ICs / PCB traces of more than 100mA / magnet component. Check HW design notice for more detail

Note 82-2: [A+G] For optmized GPS performance, please check HW design notice for Sensor selection guide

Note 82-3: [A+G] MUST use SPI for optmized sensor hub performance **DO NOT USE i2C**

Note 82-4: [A+G] Suggest choose sensor support FIFO watermark interrupt otherwise we cannot support Hifi-sensor, daydream VR. And Sensor-location accuracy will become worse.

Note 82-5: [Baro] Reserve Baro sensor for LPPe feautre (Must for North America Operator / NA SKU)

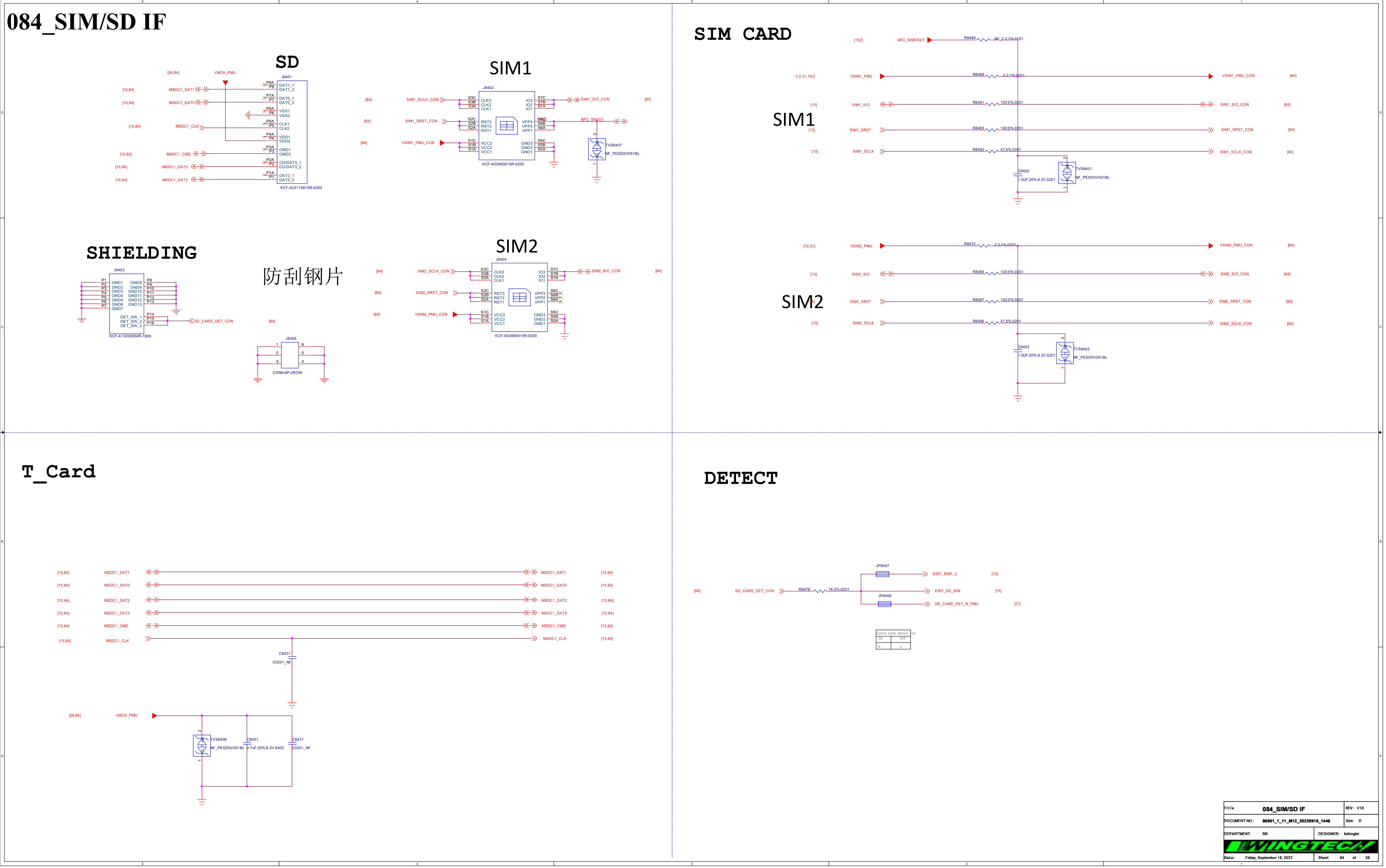
Note 82-6: DO NOT share Sensor hub i2C to other non-SCP device

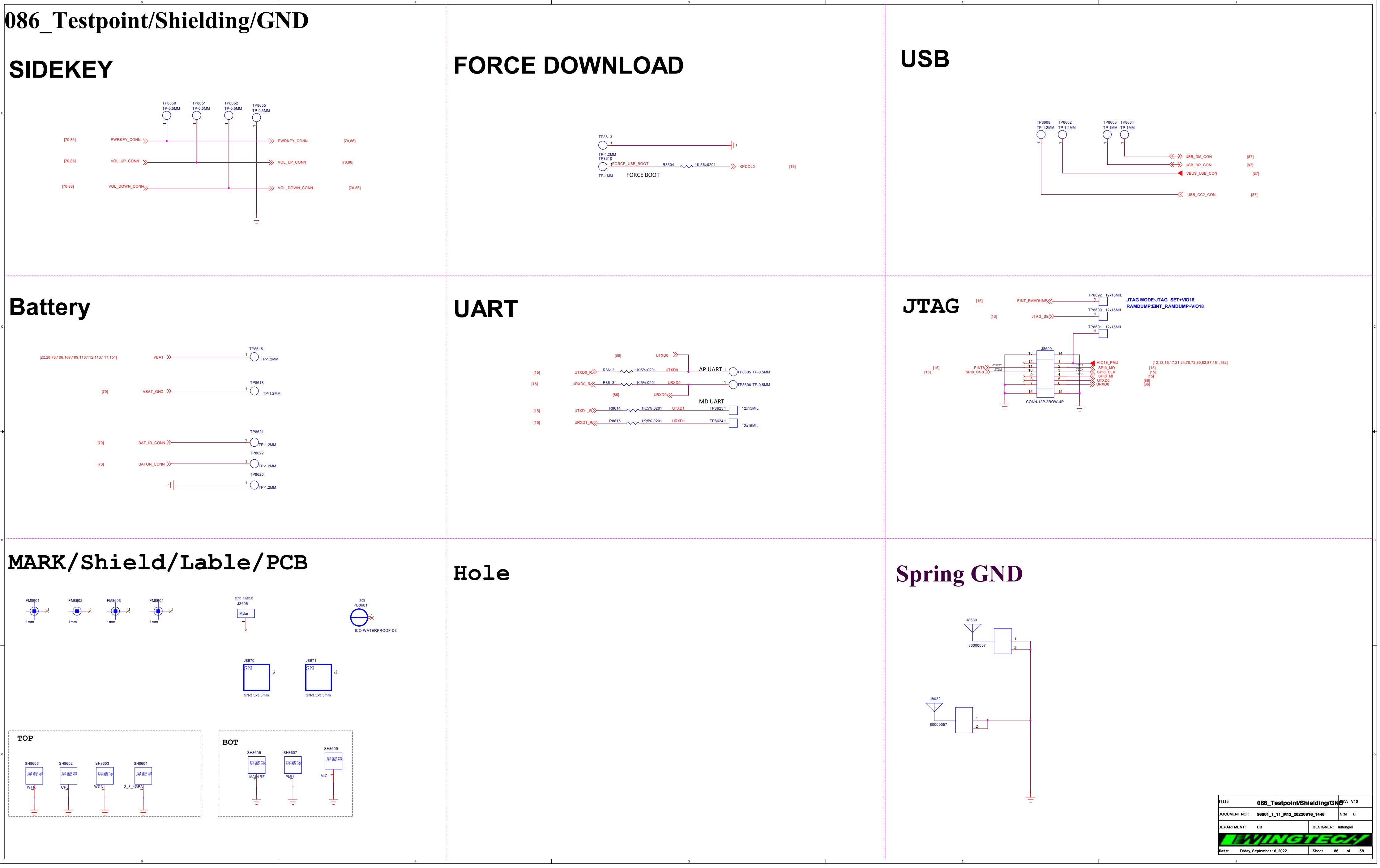
Note 82-7: Interrupt pin of MEMS sensor must be assign to ball EINT[12:0]

Vendor	Manufacture PN	Slave Address (7-bit)	I2C Address(8-bit)	Remark
Sensortek	STK33562	0X46	Read:0X8C Write:0X8D	

Title	082_SENSORS	REV: V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size D
DEPARTMENT:	BB	DESIGNER: liufenglei
Date:	Friday, September 16, 2022	Sheet 82 of 57

WINGTECH


084_SIM/SD IF
SD
J8401
KCF-A03110815R-0300
VMCH_PMU
MSDC1_DAT1
MSDC1_DAT0
MSDC1_CLK
MSDC1_CMD
MSDC1_DAT3
MSDC1_DAT2
DAT1_1
DAT1_2
DAT0_1
DAT0_2
VSS1
VSS2
CLK1
CLK2
VDD1
VDD2
CMD1
CMD2
CD/DAT3_1
CD/DAT3_2
DAT2_1
DAT2_2
SIM1
J8402
KCF-A03990615R-0200
SIM1_SCLK_CON
SIM1_SRST_CON
VSIM1_PMU_CON
SIM1_SIO_CON
NFC_SWIO1
TVS8407
NF_PESD5V0S1BL
SIM2
J8404
KCF-A03990615R-0200
SIM2_SCLK_CON
SIM2_SRST_CON
VSIM2_PMU_CON
SIM2_SIO_CON
SHIELDING
J8403
KCF-A13000004R-1000
SD_CARD_DET_CON
防刮钢片
J8405
CONN-6P-2ROW
SIM CARD
NFC_SIMVDD1
R8469
NF_2.2;1%;0201
VSIM1_PMU
R8468
2.2;1%;0201
VSIM1_PMU_CON
SIM1_SIO
R8461
100;5%;0201
SIM1_SIO_CON
SIM1_SRST
R8463
SIM1_SRST_CON
SIM1_SCLK
R8462
47;5%;0201
SIM1_SCLK_CON
C8402
1.0UF;20%;6.3V;0201
TVS8401
NF_PESD5V0S1BL
VSIM2_PMU
R8470
VSIM2_PMU_CON
SIM2_SIO
R8465
SIM2_SIO_CON
SIM2_SRST
R8467
SIM2_SRST_CON
SIM2_SCLK
R8466
SIM2_SCLK_CON
C8403
TVS8402
T_Card
C8431
C0201_NF
TVS8406
C8401
4.7uF;20%;6.3V;0402
C8477
DETECT
SD_CARD_DET_CON
R8476
1K;5%;0201
JP8407
EINT_SIM1_2
EINT_SD_SIM
JP8406
SD_CARD_DET_N_PMU
Title 084_SIM/SD IF
REV: V10
DOCUMENT NO.: 96901_1_11_M12_20220916_1446
Size D
DEPARTMENT: BB
DESIGNER: liufenglei
WINGTECH
Date: Friday, September 16, 2022
Sheet 84 of 38



086_Testpoint/Shielding/GND
SIDEKEY
TP8650 TP-0.5MM
TP8651 TP-0.5MM
TP8652 TP-0.5MM
TP8655 TP-0.5MM
[70,86] PWRKEY_CONN
[70,86] VOL_UP_CONN
[70,86] VOL_DOWN_CONN
FORCE DOWNLOAD
TP8613 TP-1.2MM
TP8610 TP-1MM
FORCE_USB_BOOT
R8604 1K;5%;0201
KPCOL0 [15]
FORCE BOOT
USB
TP8608 TP-1.2MM
TP8602 TP-1.2MM
TP8603 TP-1MM
TP8604 TP-1MM
USB_DM_CON [87]
USB_DP_CON [87]
VBUS_USB_CON [87]
USB_CC2_CON [87]
Battery
[22,28,70,106,107,109,110,112,113,117,151] VBAT
TP8615 TP-1.2MM
[70] VBAT_GND
TP8618 TP-1.2MM
[70] BAT_ID_CONN
TP8621 TP-1.2MM
[70] BATON_CONN
TP8622 TP-1.2MM
TP8620 TP-1.2MM
UART
[86] UTXD0
[15] UTXD0_R R8612 1K;5%;0201 UTXD0
AP UART
TP8605 TP-0.5MM
[15] URXD0_R R8613 1K;5%;0201 URXD0
TP8606 TP-0.5MM
[86] URXD0
MD UART
[15] UTXD1_R R8614 1K;5%;0201 UTXD1 TP8623 12x15MIL
[15] URXD1_R R8615 1K;5%;0201 URXD1 TP8624 12x15MIL
JTAG
[15] EINT_RAMDUMP TP8692 12x15MIL
[13] JTAG_SET TP8690 12x15MIL
TP8691 12x15MIL
JTAG MODE:JTAG_SET+VIO18
RAMDUMP:EINT_RAMDUMP+VIO18
J8699
CONN-12P-2ROW-4P
[15] EINT8
[15] SPI0_CSB
VIO18_PMU [12,13,15,17,21,24,70,72,80,82,87,151,152]
SPI0_MO [15]
SPI0_CLK [15]
SPI0_MI [15]
UTXD0 [86]
URXD0 [86]
MARK/Shield/Lable/PCB
FM8601 1mm
FM8602 1mm
FM8603 1mm
FM8604 1mm
J8605 Mylar
PCB PB8601
ICO-WATERPROOF-D3
J8670 SN SN-3.5x3.5mm
J8671 SN SN-3.5x3.5mm
TOP
SH8605 屏蔽罩 WTR
SH8602 屏蔽罩 CPU
SH8603 屏蔽罩 WCN
SH8604 屏蔽罩 2_3_4GPA
BOT
SH8606 屏蔽罩 MAIN RF
SH8607 屏蔽罩 PMU
SH8609 屏蔽罩 MIC
Hole
Spring GND
J8630 80000057
J8632 80000057
Title 086_Testpoint/Shielding/GND
REV: V10
DOCUMENT NO.: 96901_1_11_M12_20220916_1446
Size D
DEPARTMENT: BB
DESIGNER: liufenglei
WINGTECH
Date: Friday, September 16, 2022
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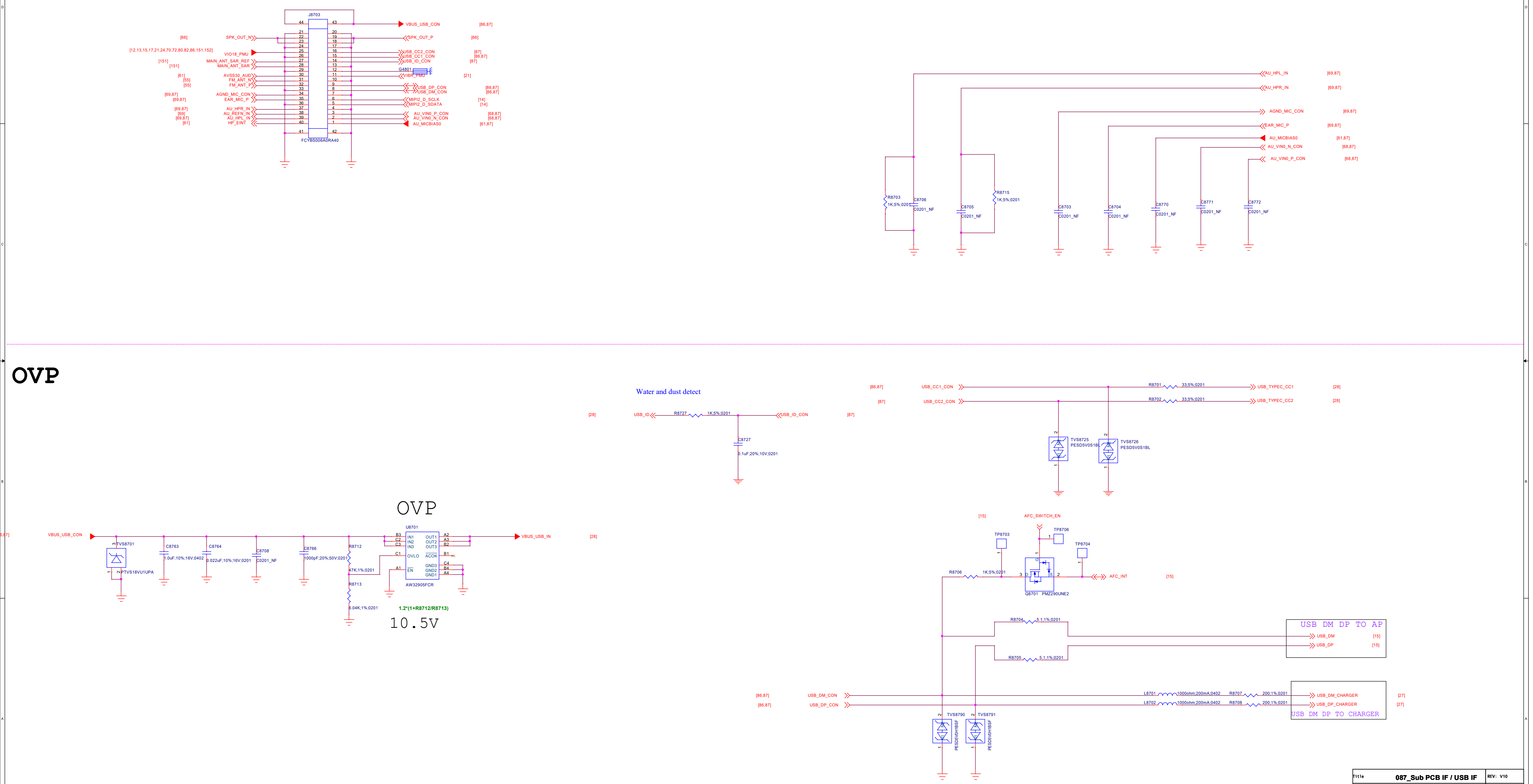
087_Sub PCB IF / USB IF

Customer P/N

AUDIO CAP

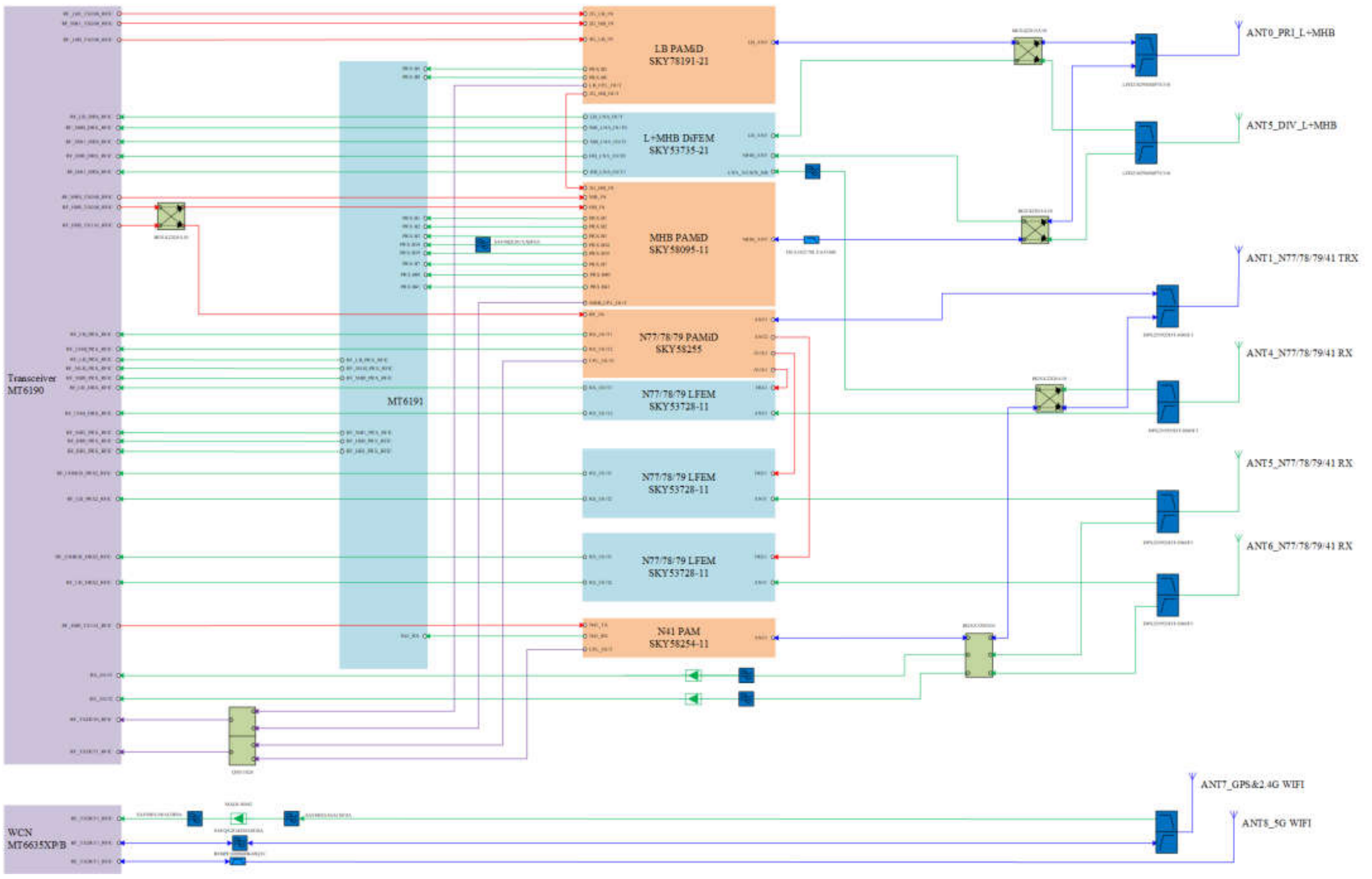


Title	087_Sub PCB IF / USB IF		REV: V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446		Size D
DEPARTMENT:	BB	DESIGNER:	liufenglei
WINGTECH			
Date:	Friday, September 16, 2022	Sheet	87 of 38

101.RF_Block

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

RF Block Diagram



Title	101.RF_Block	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	Custom
DEPARTMENT:	RF	DESIGNER:	Xue Wei
Date:	Friday, September 16, 2022	Sheet	101 of 152

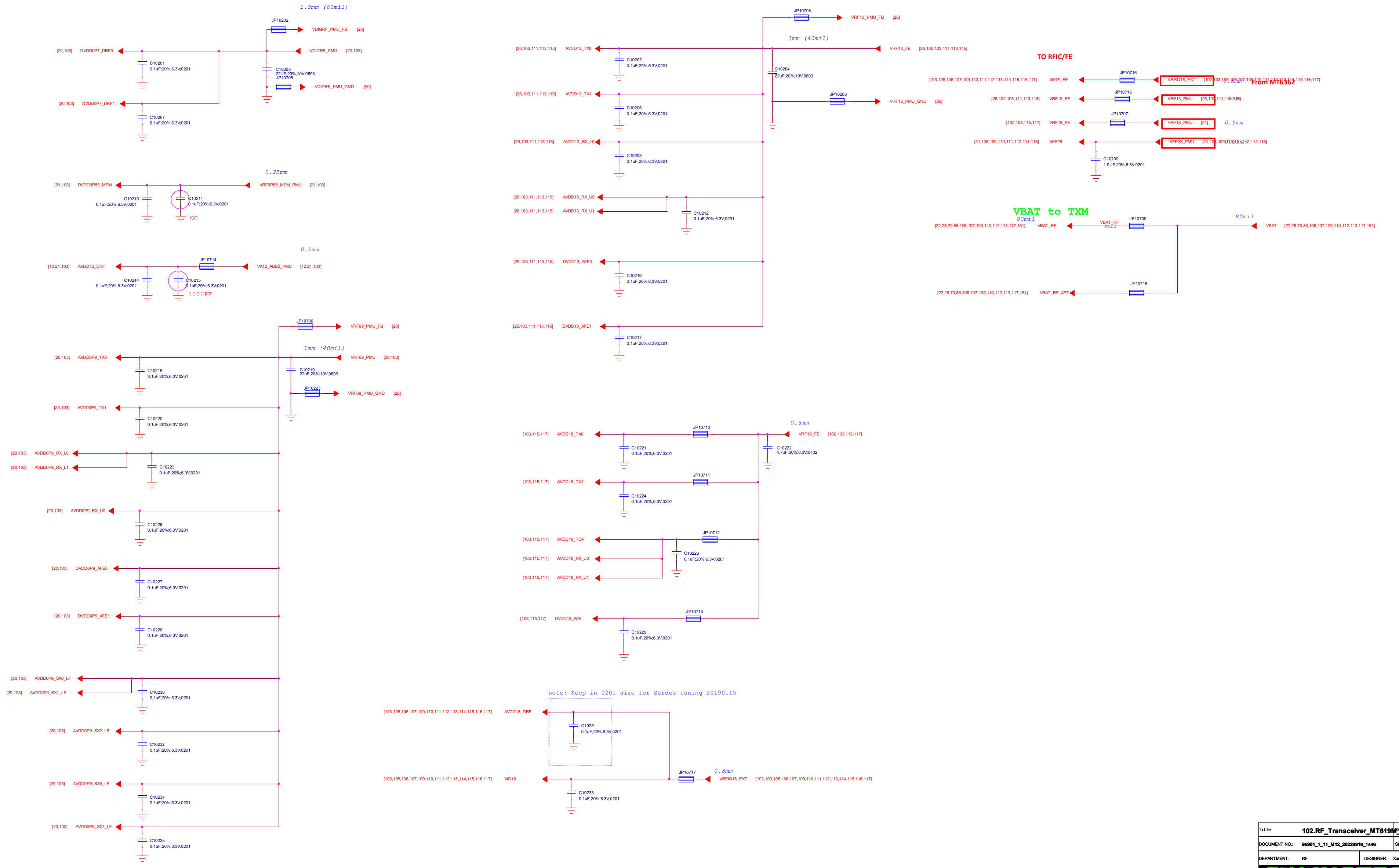
WINGTECH

VERSION	EDITED BY	LAST EDIT DATE
1	LAST_EDITOR	2-23-2007_10:06

102.RF_Transceiver_MT6190M_Power

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

Power Supply of Transceiver



Title	102.RF_Transceiver_MT6190M_Power
DOCUMENT NO.:	96901_1_11_M12_20220916_1446
Size	Custom
DEPARTMENT:	RF
DESIGNER:	Xue Wei
Date:	Friday, September 16, 2022
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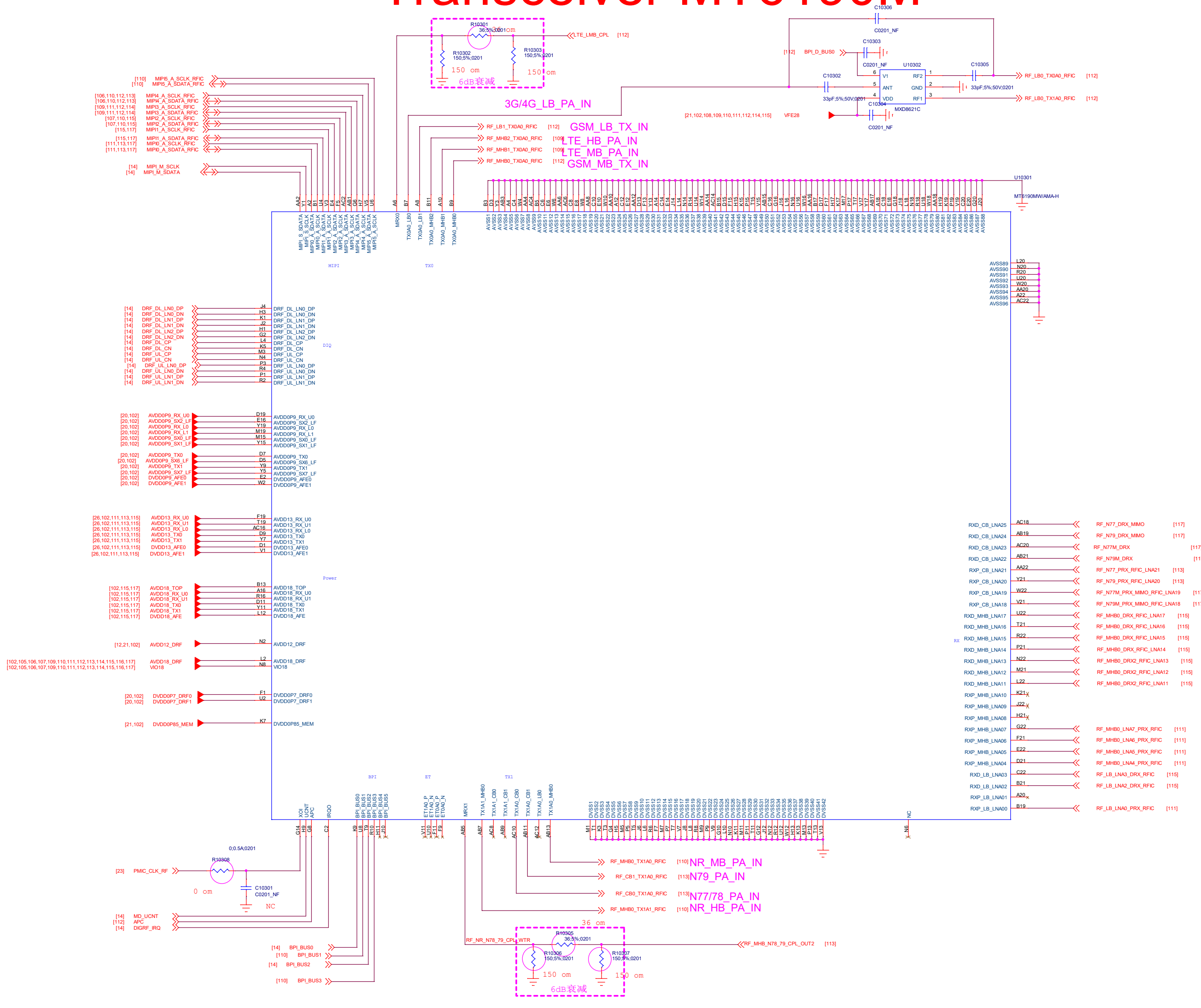
1	LAST_EDITOR	2-23-2007_10:06

103.RF_Transceiver_MT6190M

Transceiver MT6190M

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



Title	103.RF_Transceiver_MT6190M	Rev	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	Custom
DEPARTMENT:	RF	DESIGNER:	Xue Wei
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1	LAST_EDITOR	2-23-2007_10:06

104.RF interface

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

Title	104.RF interface	REV: V10	
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	Custom
DEPARTMENT:	RF	DESIGNER:	Xue Wei
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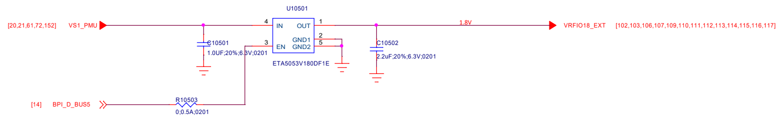
1	LAST_EDITOR	2-23-2007_10:06

105_POWER_EXT

External VRFIO18_LDO



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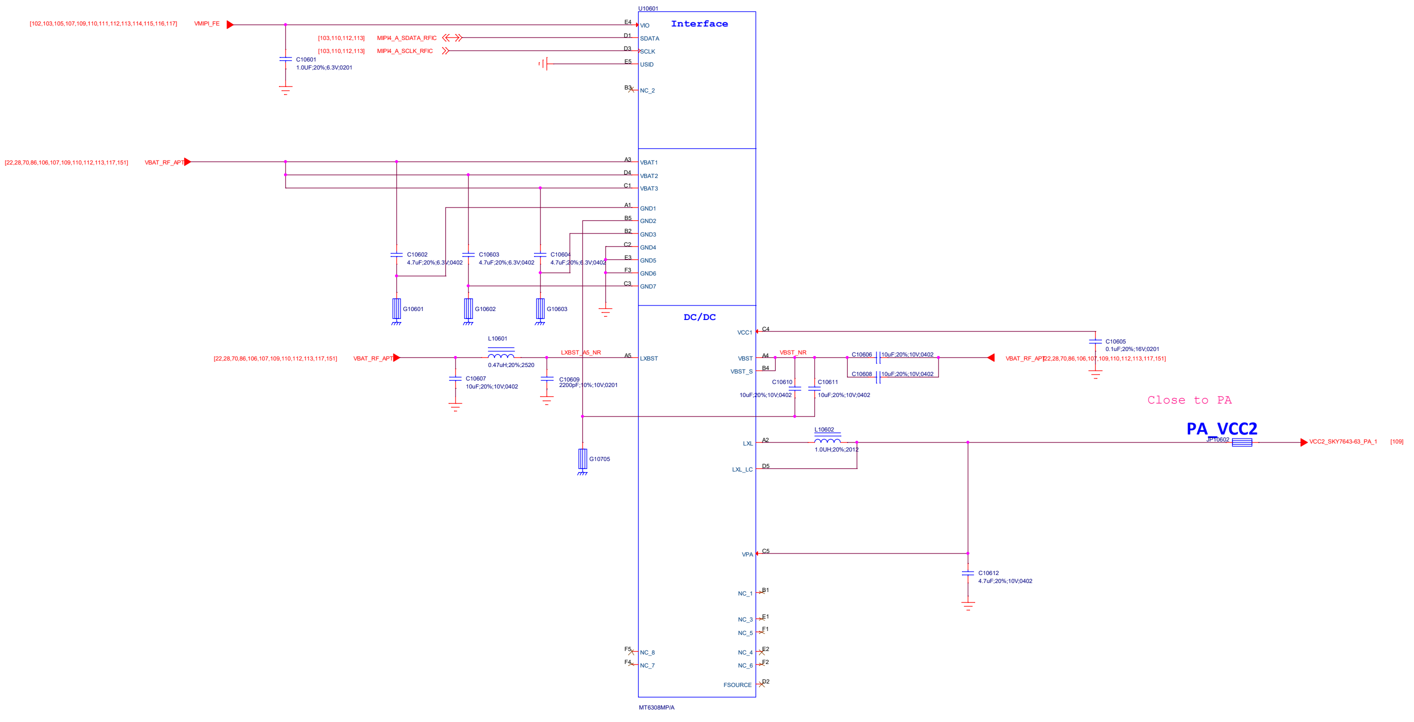
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT: BB		DESIGNER: Ning Xinwu	
Date: Friday, September 16, 2022		Sheet 105 of 152	

6308M_APT _1#

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



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DOCUMENT NO.:	96901_1_11_M12_20220916_1446		Size Custom
DEPARTMENT:	RF	DESIGNER:	Xue Wei
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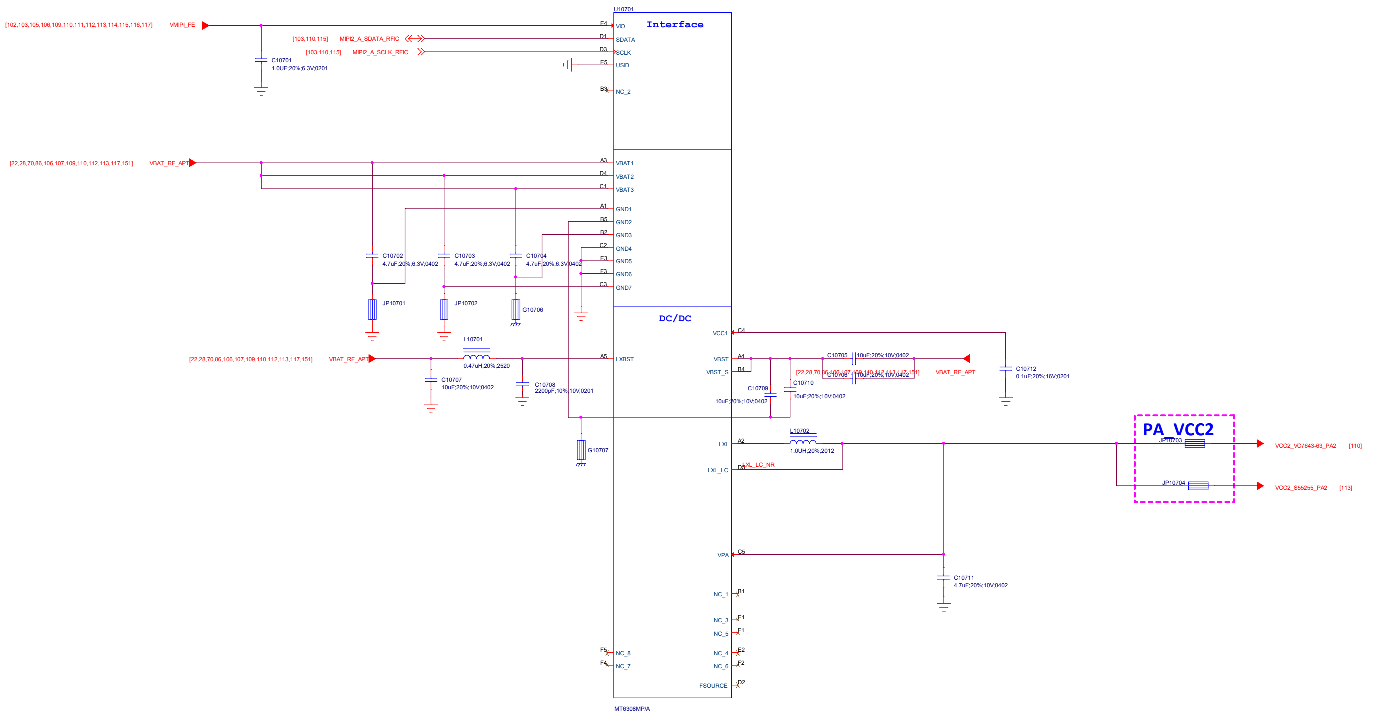
1	LAST_EDITOR	2-23-2007_10:06

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

6308M_APT _2#



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DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	Custom
DEPARTMENT: RF		DESIGNER:	Xue Wei
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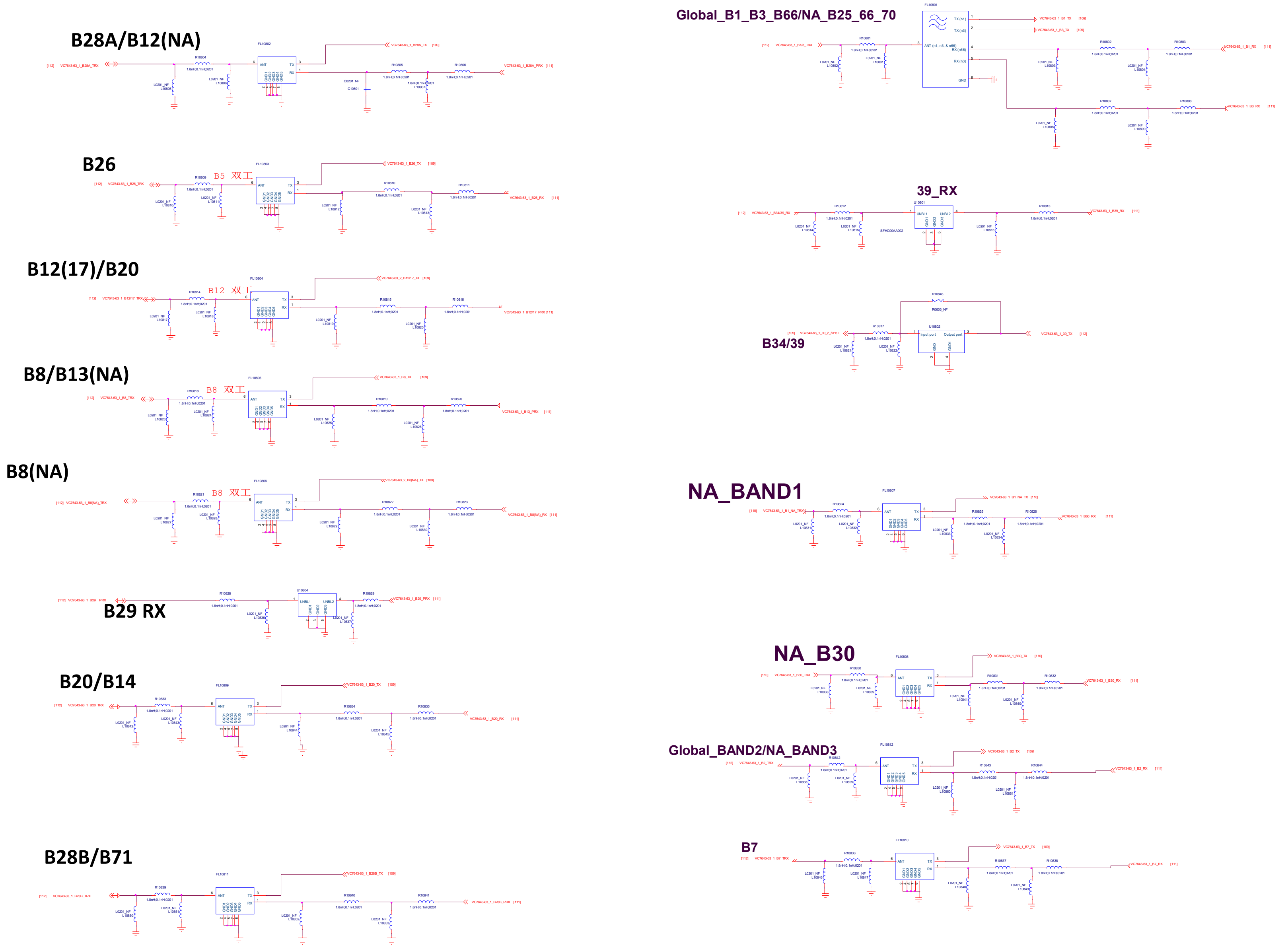
1	LAST_EDITOR	2-23-2007_10:06



Title	108.VC7643-63#1_LTE_LMHB_TRX	REV	
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	E
DEPARTMENT:	RF	DESIGNER:	Xue Wei
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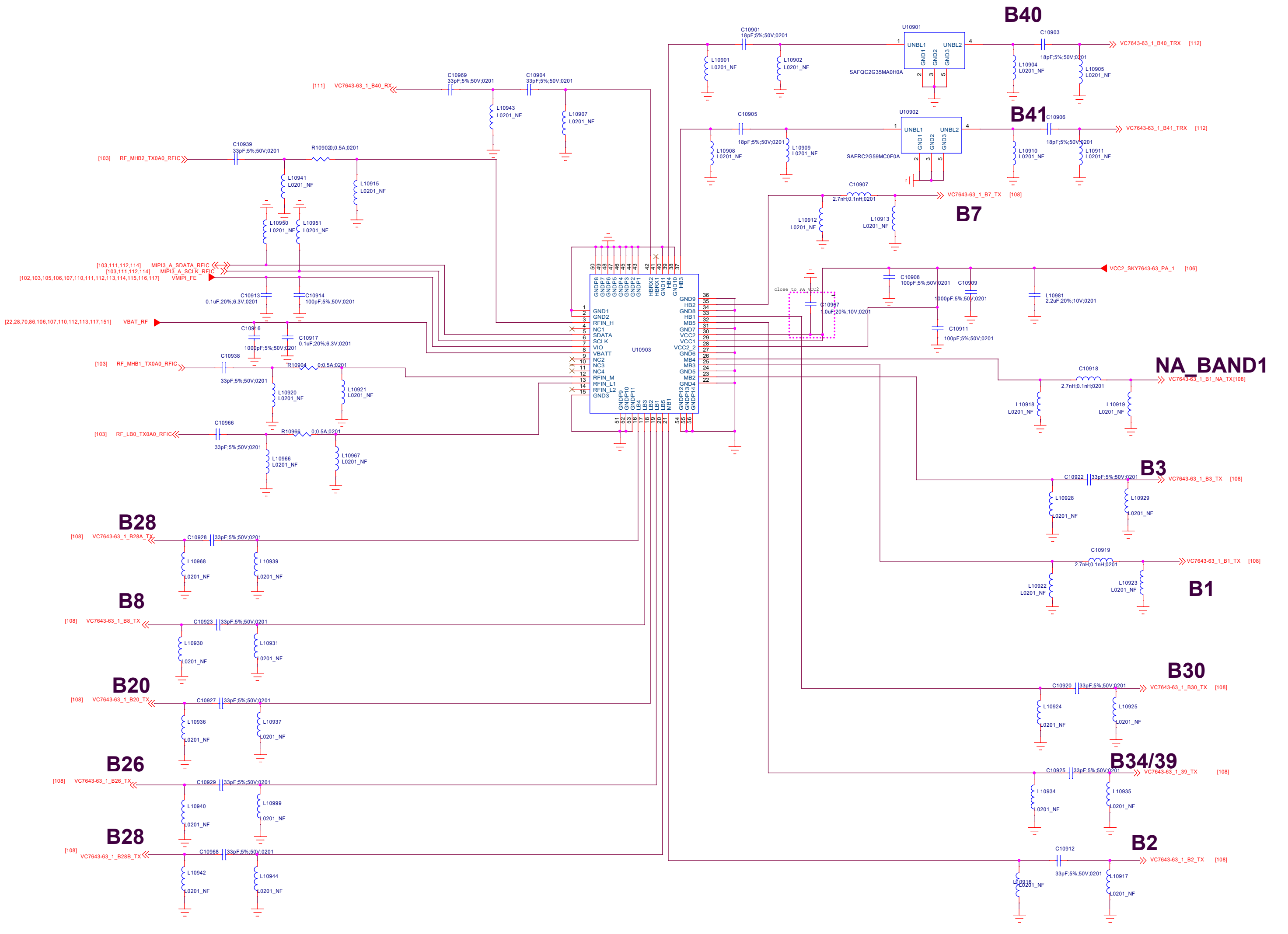
109.VC7643-62#1



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DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	RF	DESIGNER:	Xue Wei
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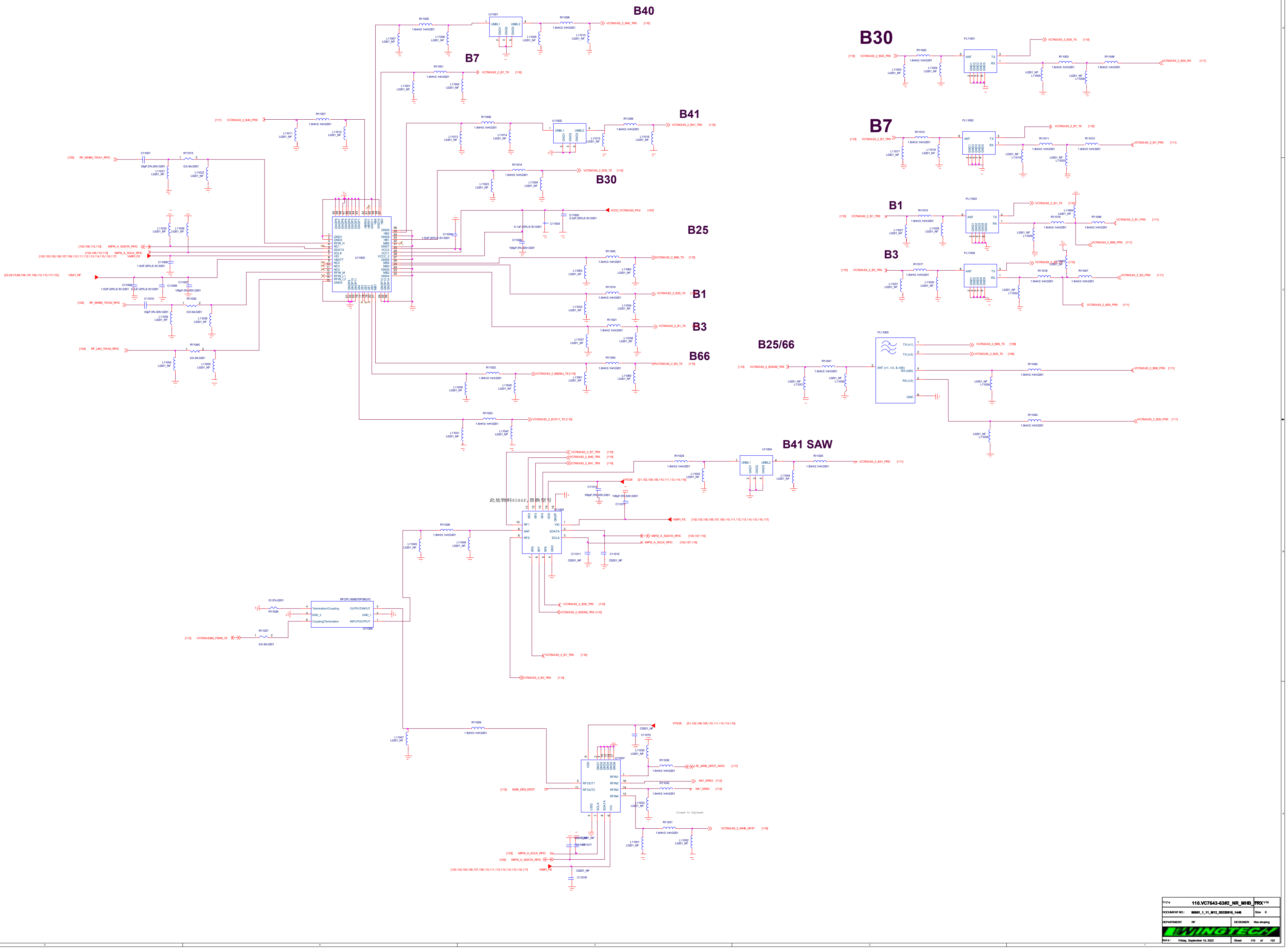
110.VC7643-63#2



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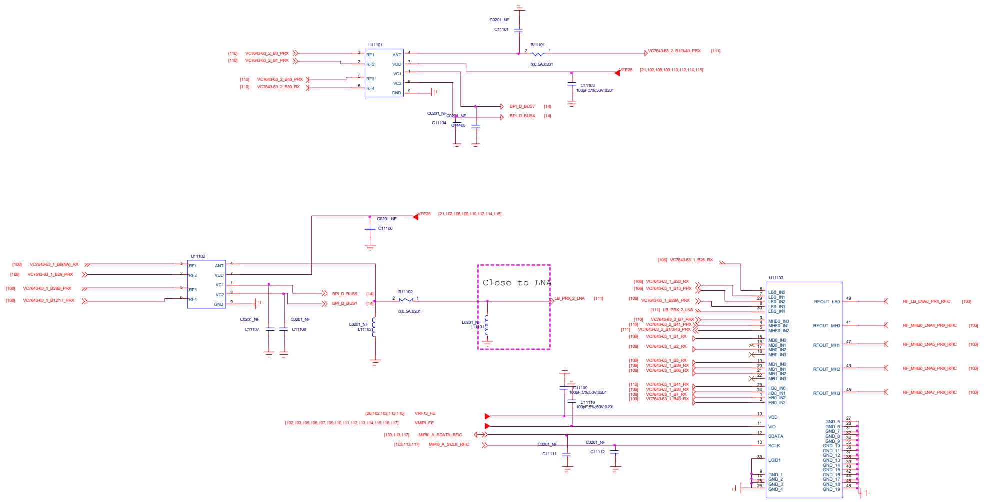

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DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	E
DEPARTMENT:	RF	DESIGNER:	Ren shuping
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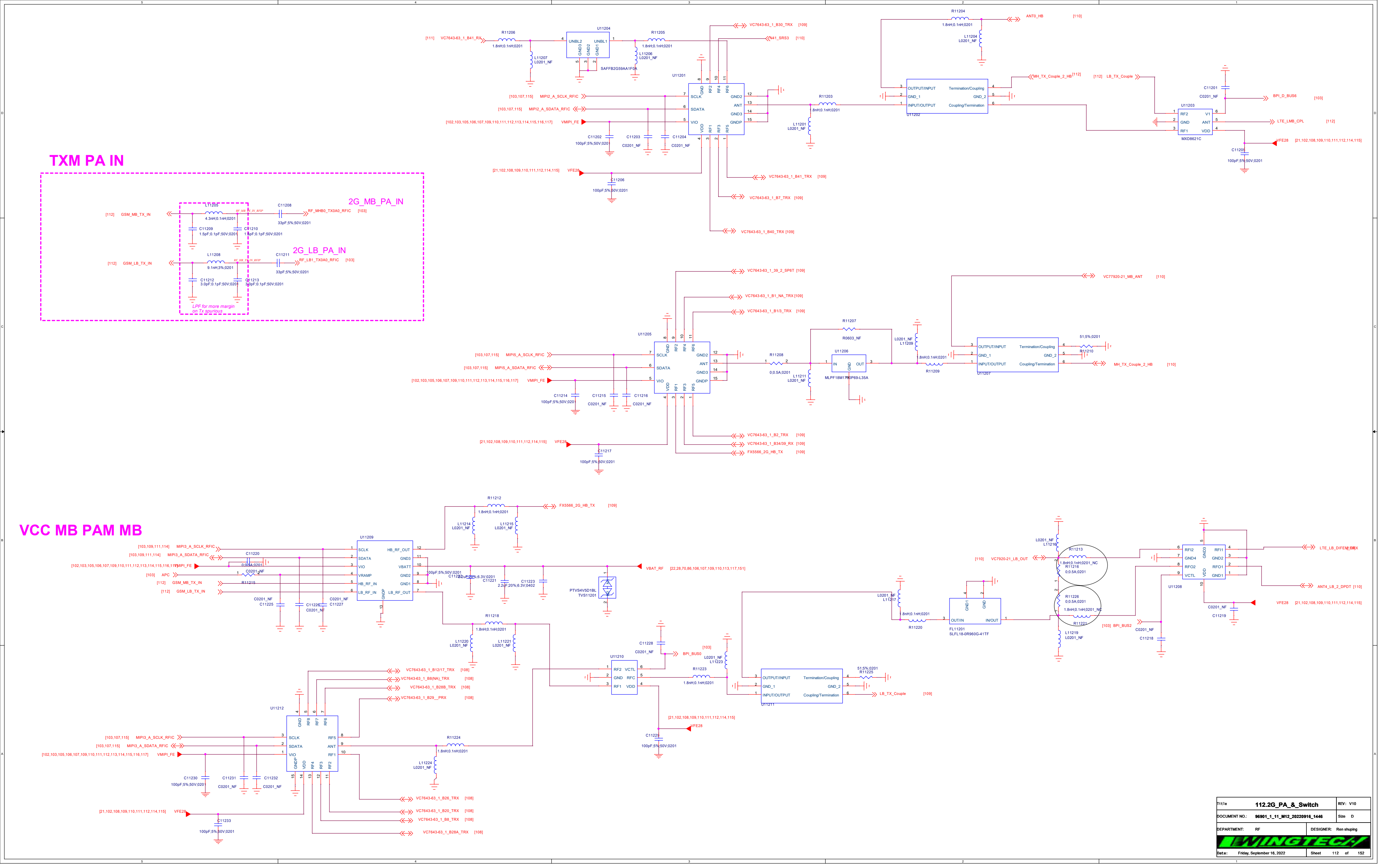

Title	112.2G_PA_&_Switch		REV: V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446		Size D
DEPARTMENT:	RF	DESIGNER:	Ren shuping
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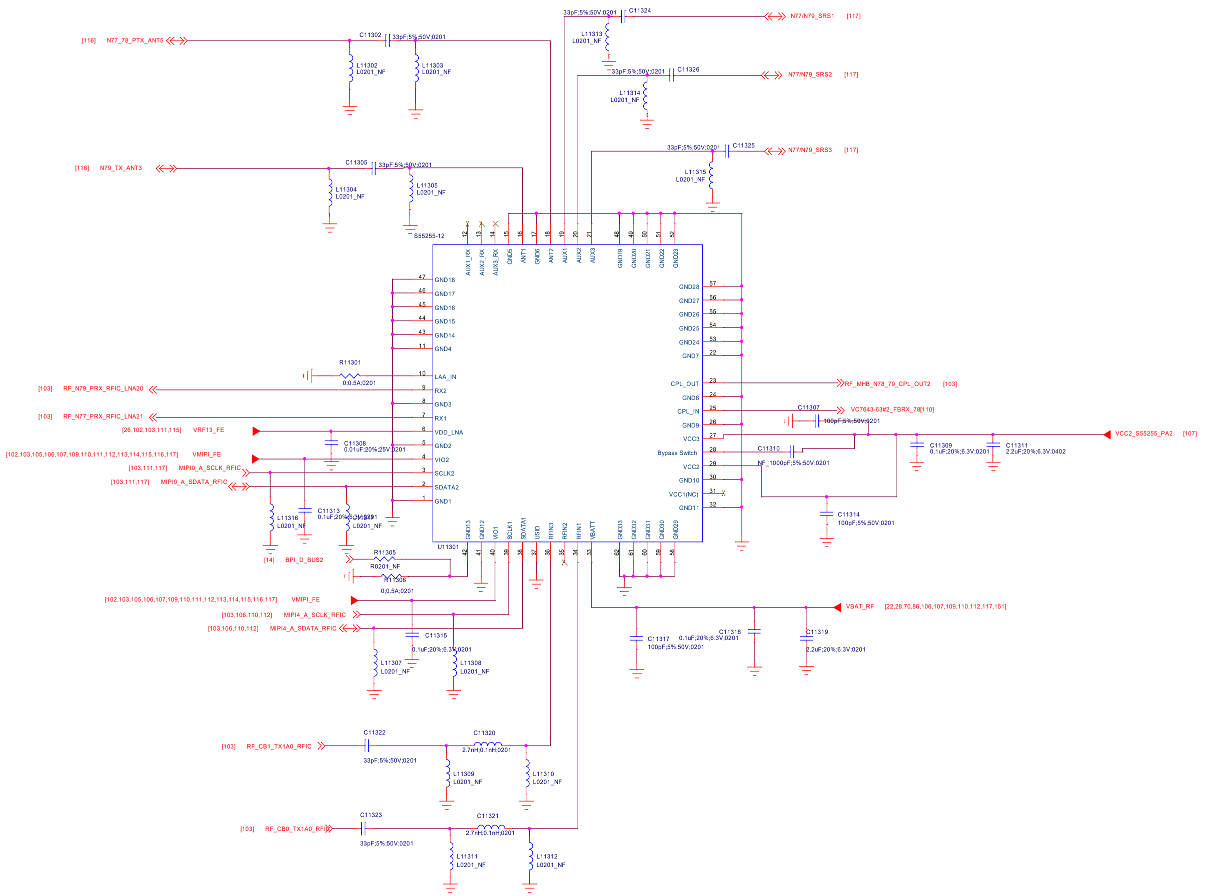
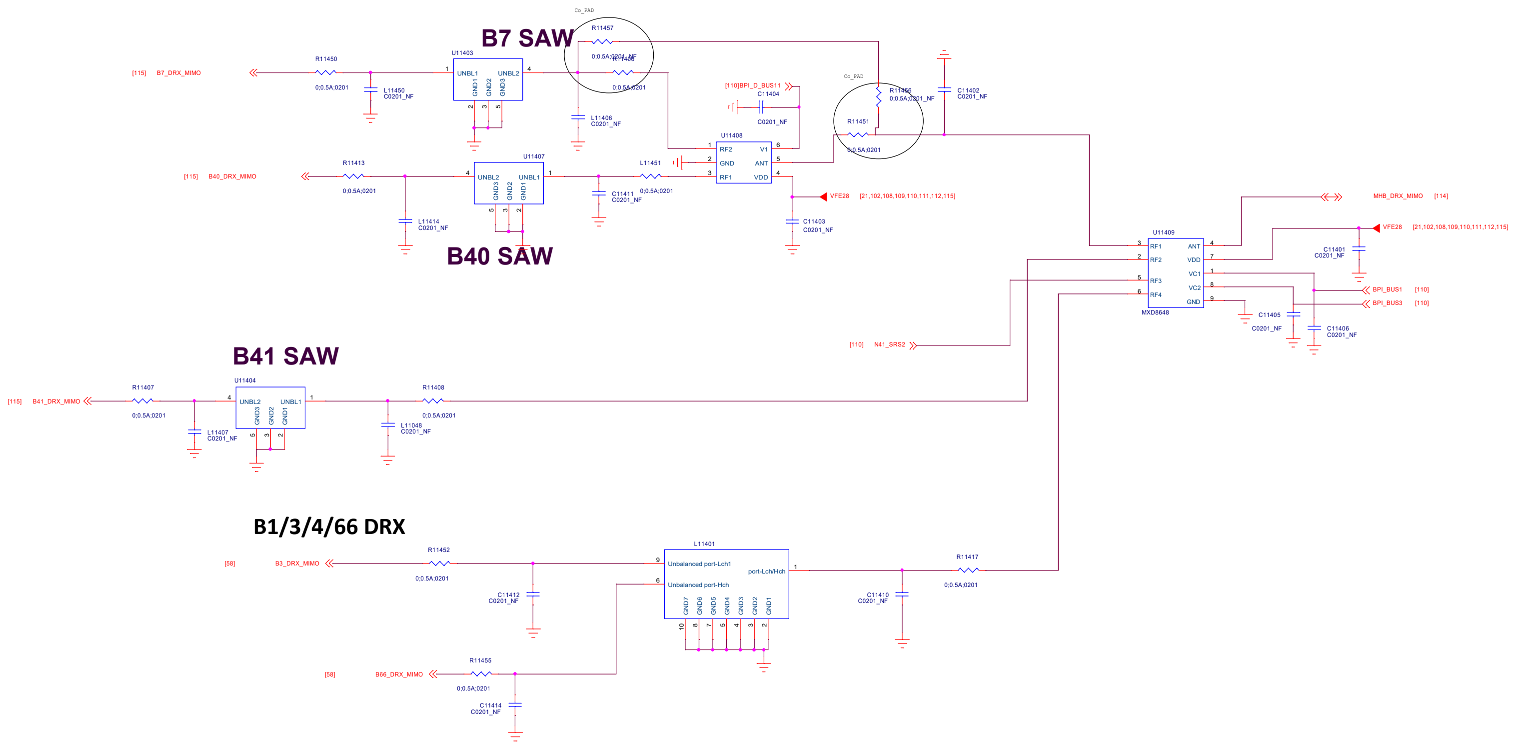
B48/N77/79_TRX



Title	113.N77/78 TRX	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	RF	DESIGNER:	Ren shuping
Date:	Friday, September 16, 2022	Sheet	113 of 152

MHB_DRX_MIMO



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DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	RF	DESIGNER:	Ren shuping
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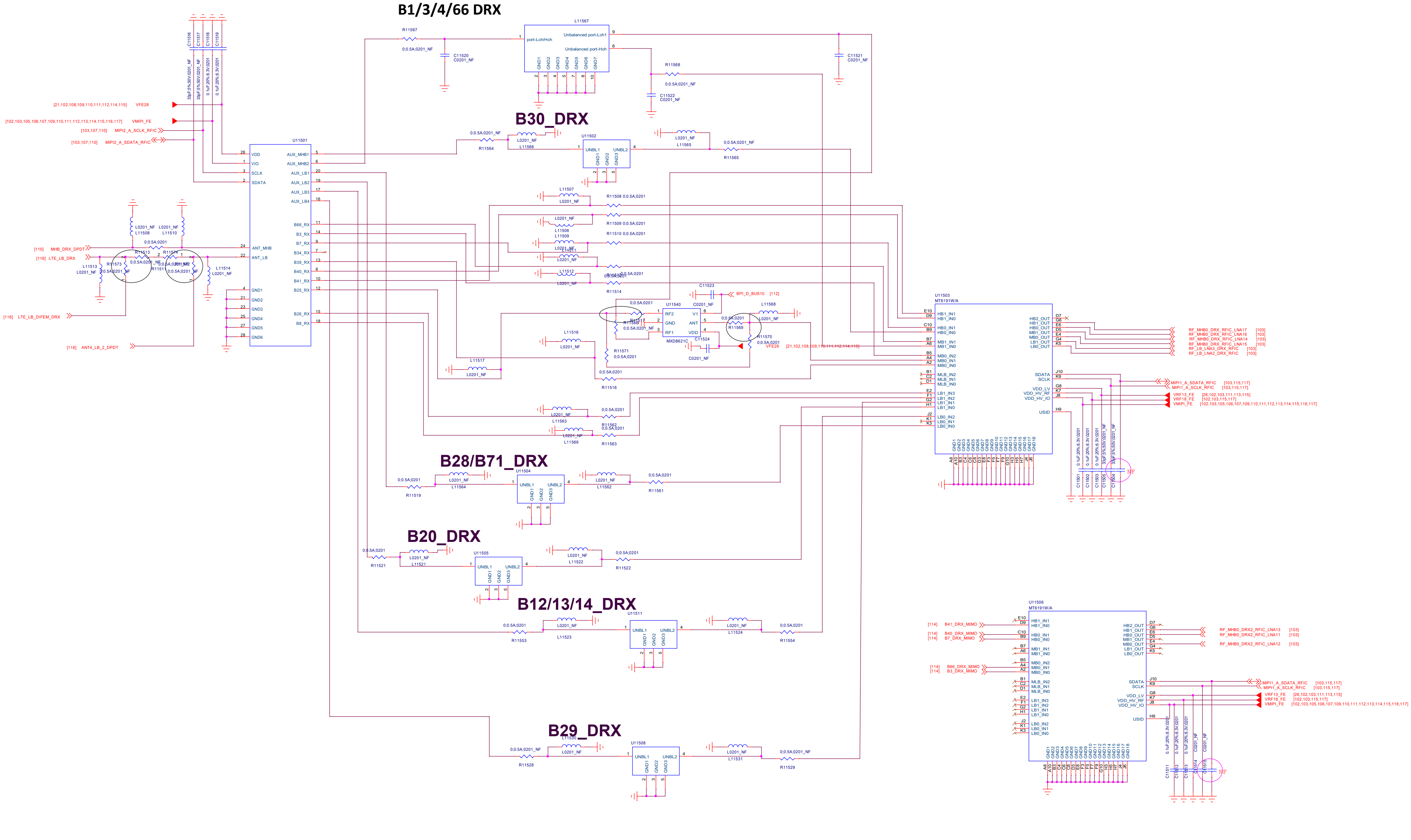

Title	115.DRX	REV: V10
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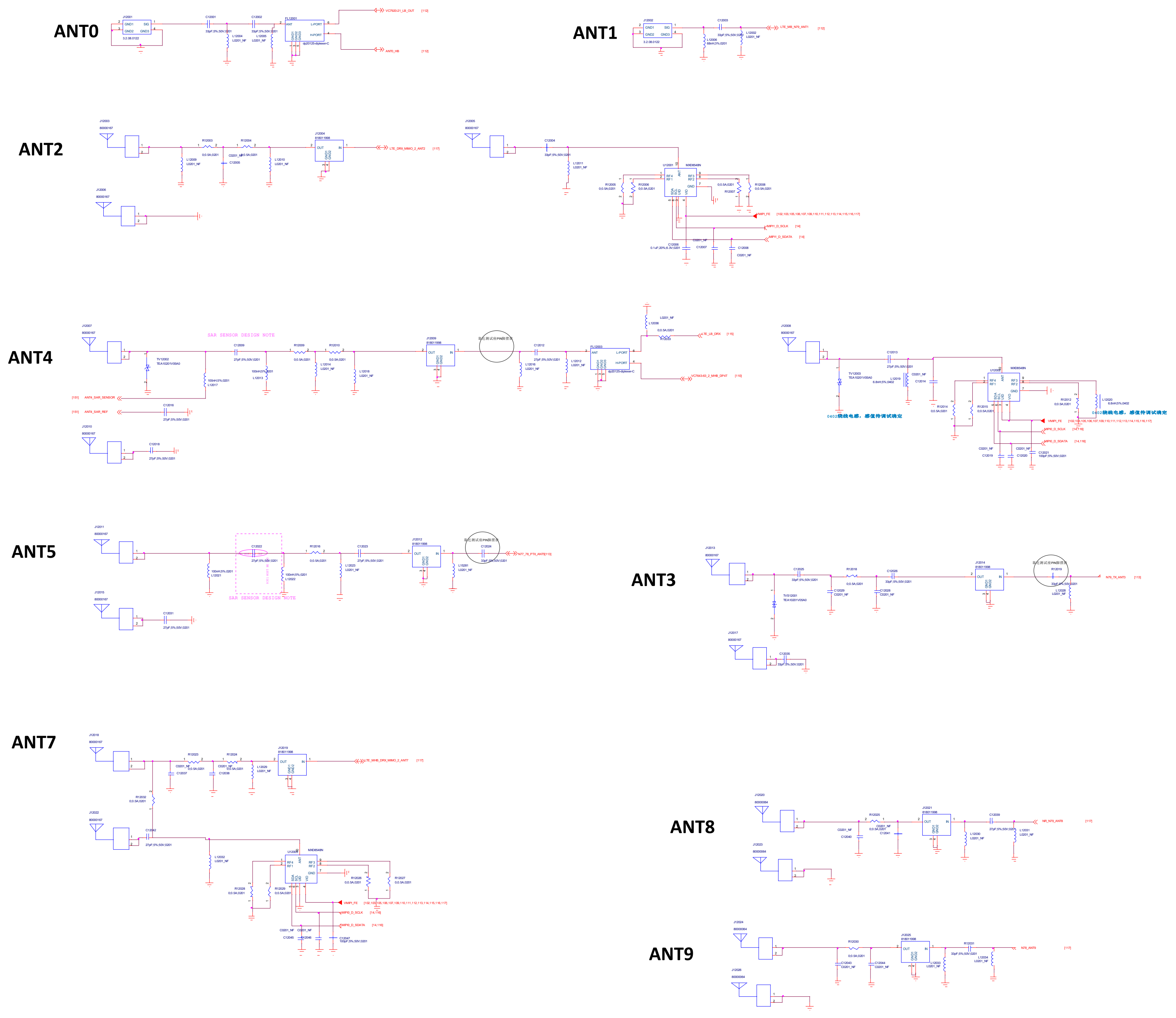
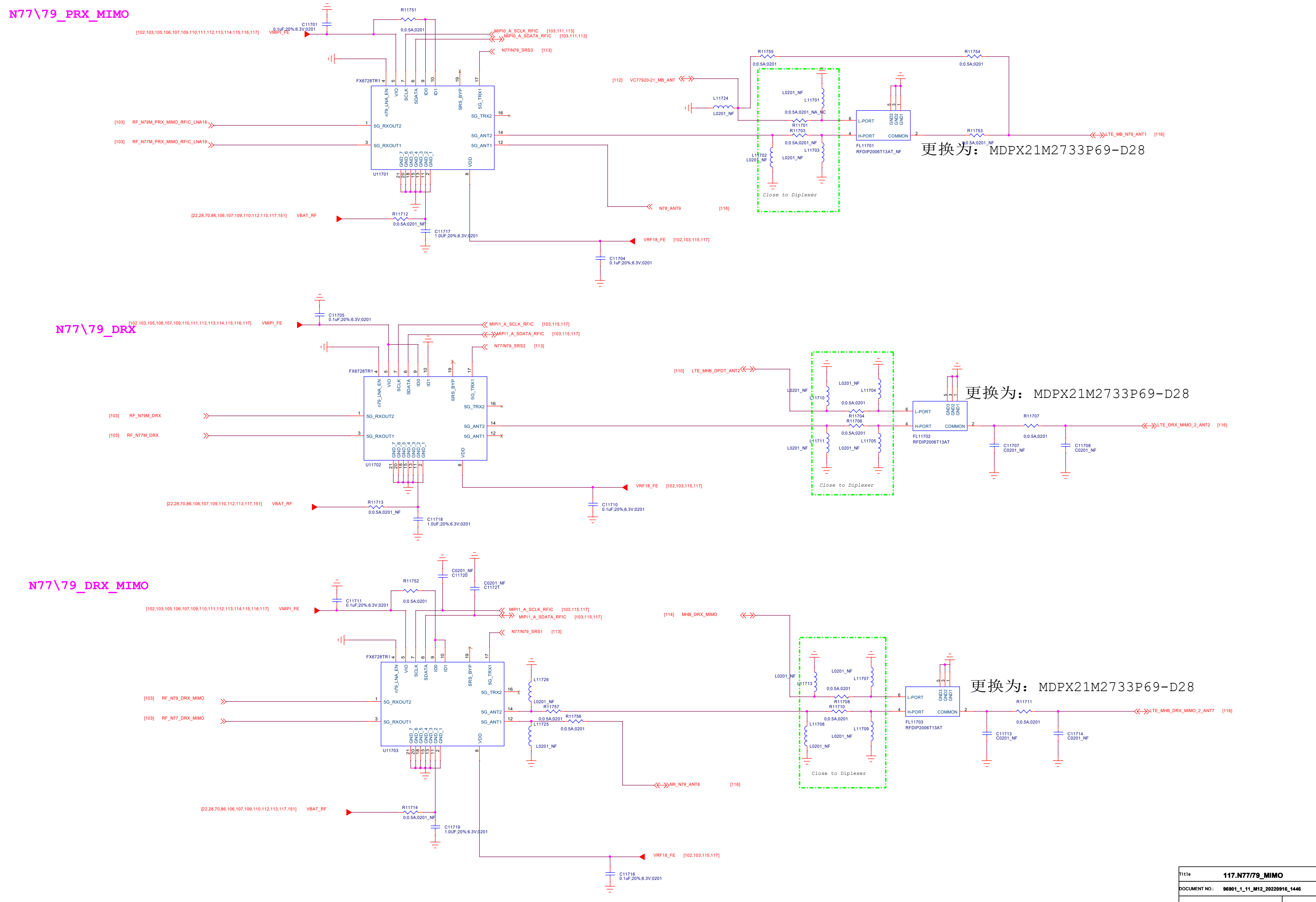
DEPARTMENT: RF	DESIGNER: Ren shuping	
Date: Friday, September 16, 2022	Sheet 115 of 152	



Title	120.ANT	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	E
DEPARTMENT:	RF	DESIGNER:	Ren shuping
Date:	Friday, September 16, 2022	Sheet	116 of 152



Title	117.N77/79_MIMO	REV:	V10
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size	D
DEPARTMENT:	RF	DESIGNER:	Ren shuping
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Title	118.XXXXXX	REV: V10	
DOCUMENT NO.:	96901_1_11_M12_20220916_1446	Size D	
DEPARTMENT:	RF	DESIGNER:	Xue Wei
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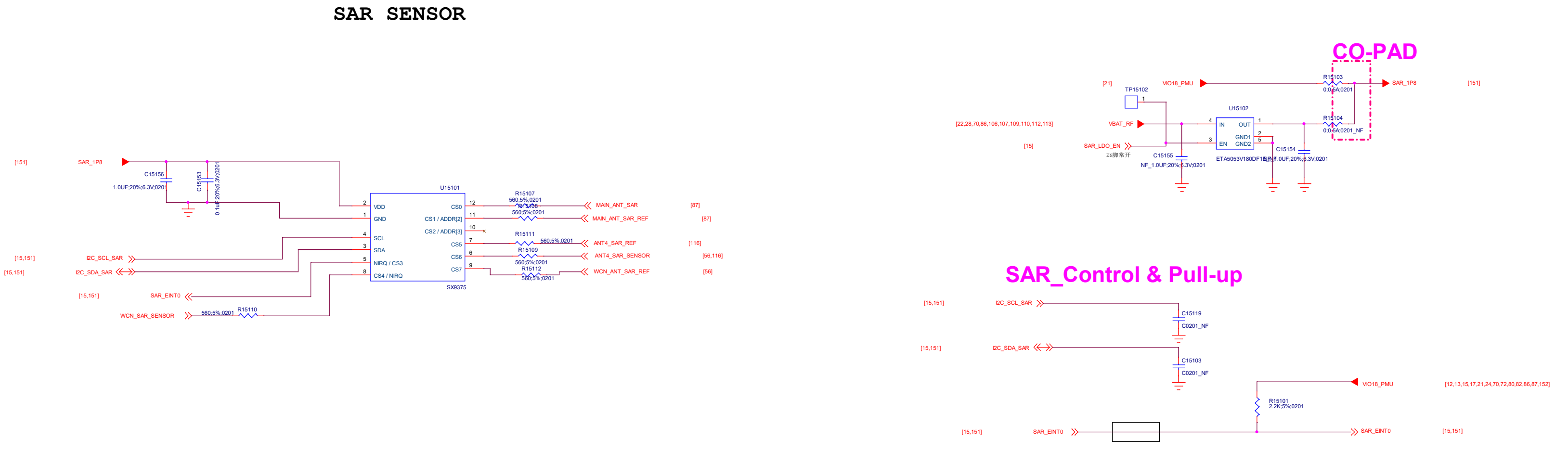
151.Sar_sensor

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



Title 151.Sar_sensor	REV: V10
DOCUMENT NO.: 96901_1_11_M12_20220916_1446	Size Custom
DEPARTMENT: WINGTECH-XA	DESIGNER: Xue Wei
Date: Friday, September 16, 2022	Sheet 151 of 67

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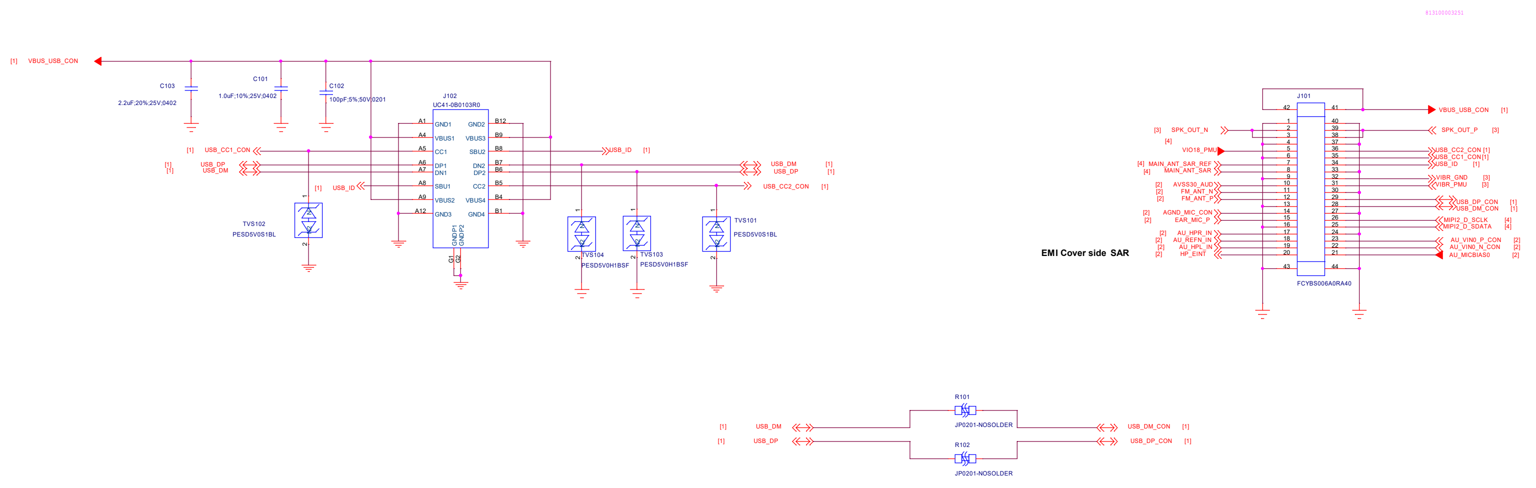
USB Connector



Title	01.USB IF	REV:	V10
DOCUMENT NO.:	96901_2_11_M12_20220914	Size	D
DEPARTMENT:	DEPARTMENT	DESIGNER:	DESIGNER
Date:	Wednesday, September 14, 2022	Sheet 1 of 4	

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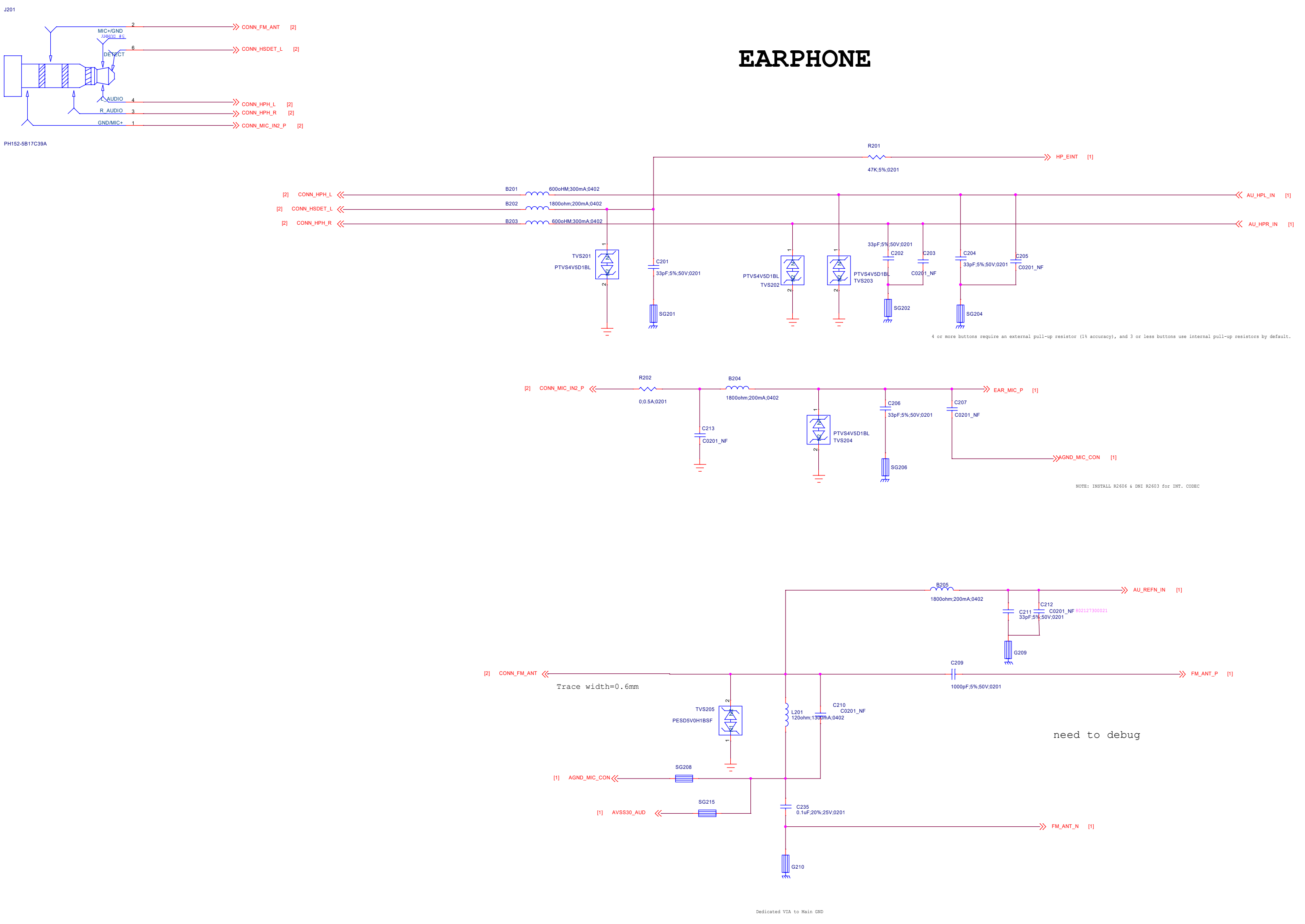
EARPHONE

Note: EURO/US compatible design,
Default US type



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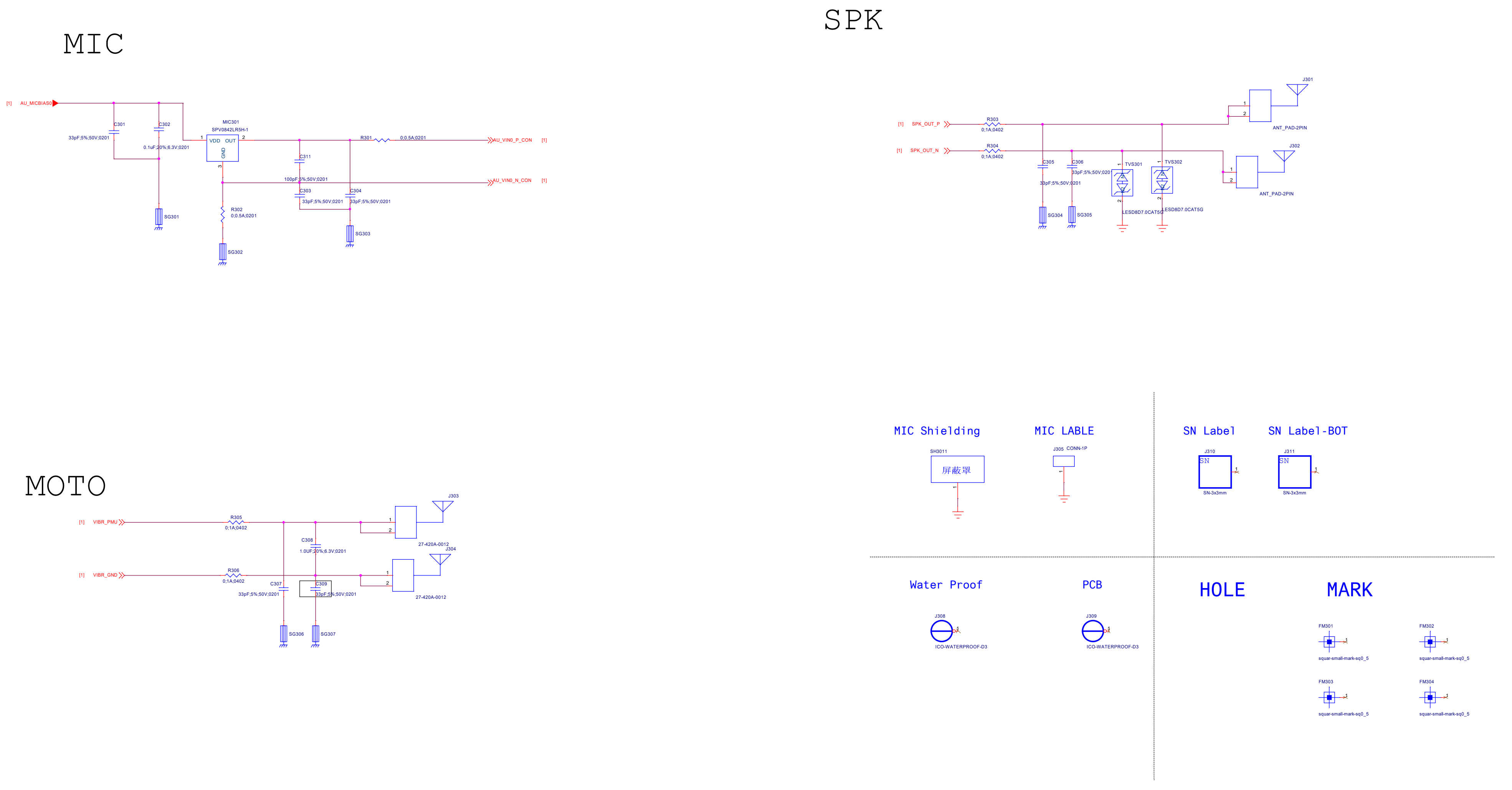
DOCUMENT NO.:	96901_2_11_M12_20220914	Size D
DEPARTMENT:	DEPARTMENT	DESIGNER: DESIGNER
	WINGTECH	
Date:	Friday, June 24, 2022	Sheet 2 of 4



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DOCUMENT NO.:	96901_2_11_M12_20220914	Size	D
DEPARTMENT:	DEPARTMENT	DESIGNER:	DESIGNER
Date:	Thursday, August 04, 2022	Sheet	3 of 4

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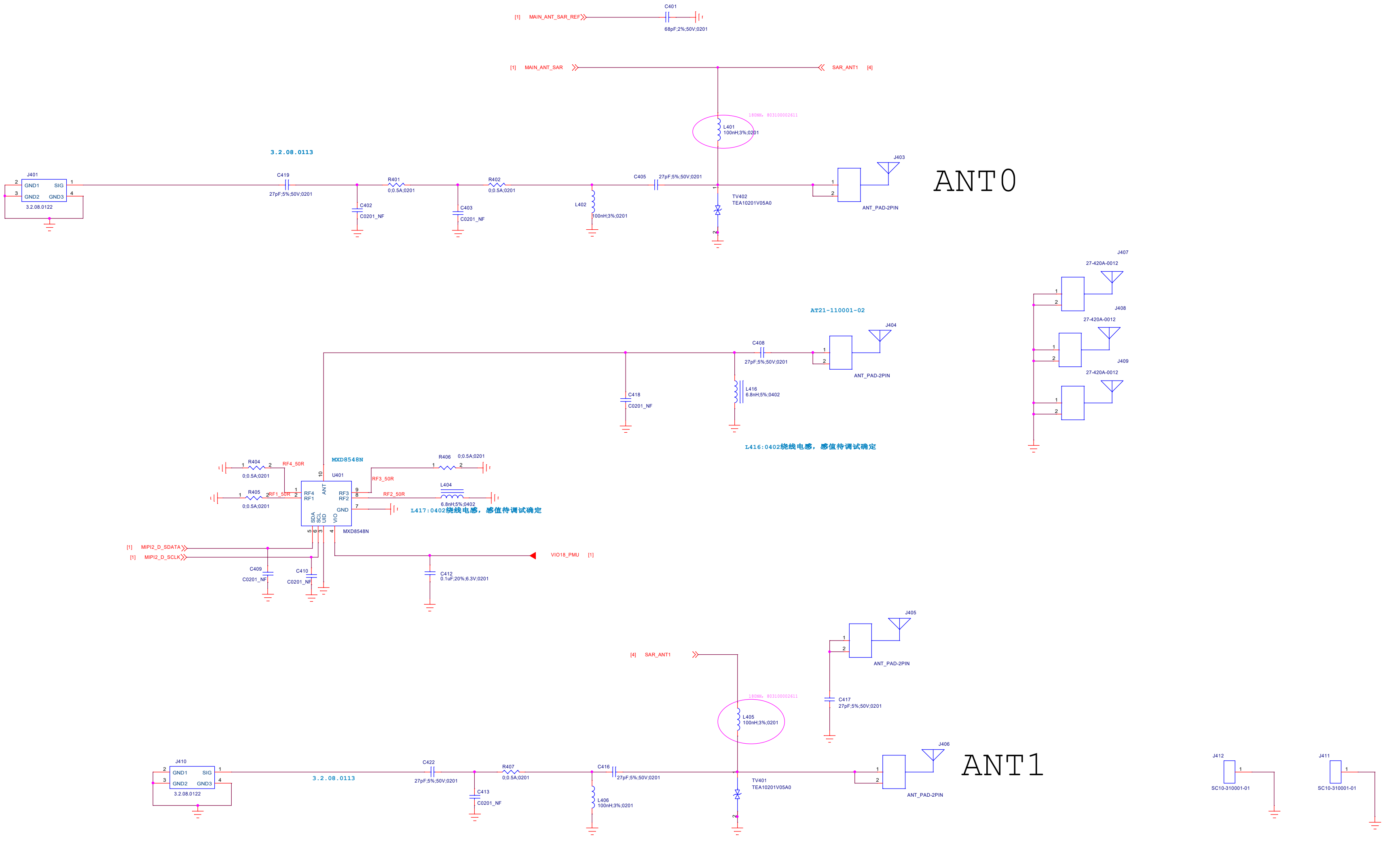

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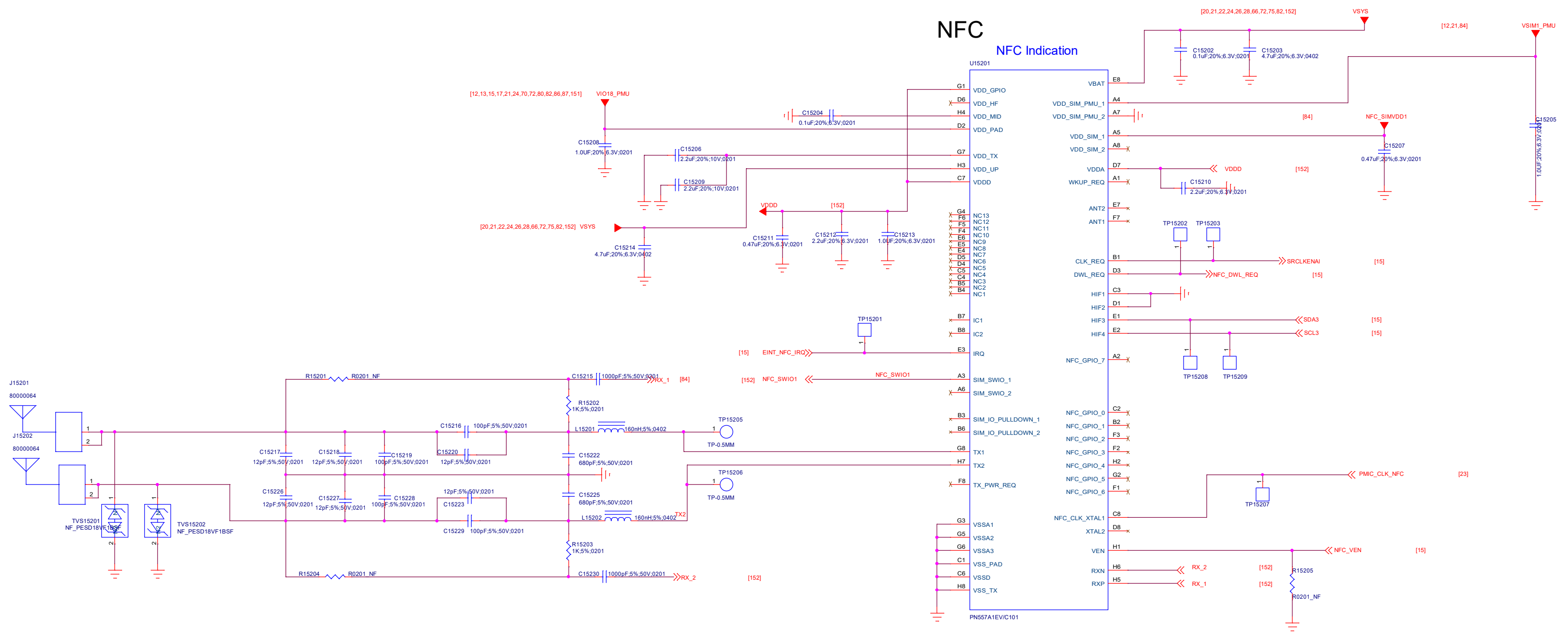
DEPARTMENT:	DEPARTMENT	DESIGNER:	DESIGNER
WINGTECH			
Date:	Tuesday, July 26, 2022	Sheet 4	of 4

NFC



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DEPARTMENT:	WINGTECH-SH	DESIGNER:	liufenglei
WINGTECH			
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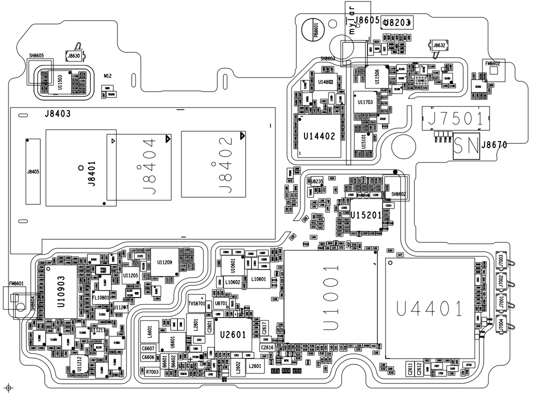

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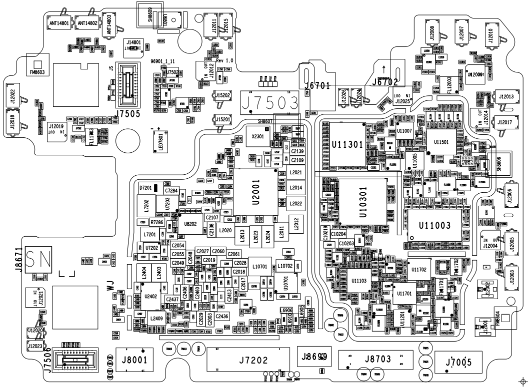
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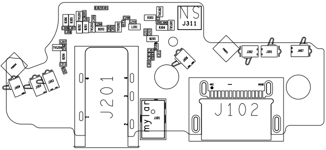

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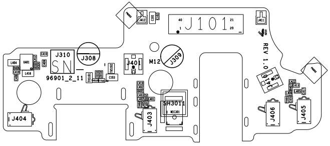

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